

Academic Council:16/11/2024

Item No:3.2

Parle Tilak Vidyalaya Association's
**MULUND COLLEGE OF COMMERCE
(AUTONOMOUS)**



Syllabus for T.Y. B.Sc. I.T.

Programme: B.Sc. I.T.

Code:

**BASED ON LEARNING OUTCOME
CURRICULUM FRAMEWORK (LOCF)**

Semester V & VI

with effect from the academic year

2025 – 2026

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**Parle Tilak Vidyalaya Association's
MULUND COLLEGE OF COMMERCE (AUTONOMOUS)**



Sr. No.	Heading	Particulars
1.	Title of the Programme	B.Sc. (Information Technology)
2.	Eligibility for Admission	S.Y.B.Sc. I.T.
3.	Passing Marks	40%
4.	Ordinances / Regulations (if, any)	As applicable for all B.Sc. Programmes, University of Mumbai
5.	Number of years / Semesters	Three years – Six Semesters
6.	Level	P.G. / U.G. / Diploma / Certificate (Strike out which is not applicable)
7.	Pattern	Yearly / Semester, Choice Based (Strike out which is not applicable)
8.	Status	New / Revised
9.	To be implemented from Academic year	From the Academic Year <u>2025 – 2026</u>

Date: November 16, 2024
Name of the BoS Chairperson

Signature: 
Dr. Hiren Dand
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Semester – 5			
Course Code	Course Type	Course Title	Credits
MCCSCT403	Major	DevOps	4
MCCSCT309	Major	ASP.NET Core	4
MCCSCT310	Major	Big Data and NOSQL	4
MCCSCT404	(Choose any one)	Enterprise Java	
MCCSCT306	Minor (Choose any one)	Internet of Things	4
MCCSCT311	VSEC	Artificial Intelligence	4
MCCSCT307	(Choose any one)	Virtual Reality & Augmented Reality	
MCCOJTSCT302	RP / OJT	OJT	2
MCCSCTPRJ401		Project	
Total Credits			22

Semester – 6			
Course Code	Course Type	Course Title	Credits
MCCSCT405	Major	Computer Security	4
MCCSCT406	Major	Business Intelligence	4
MCCSCT408	Major	Advanced Mobile Programming Practical	2
MCCSCT313	Minor (Choose any one)	Principles of GIS	2
MCCSCT314		Cloud Computing Fundamentals	
MCCSCT315	Minor (Choose any one)	IT Service Management	2
MCCSCT316		Cyber Laws and Patents	
MCCSCT407	VSEC (Choose any one)	EARN	4
MCCSCT312		Linux Administration	
MCCSCTPRJ402	RP	CAPSTONE Project	4
Total Credits			22

VSEC: Vocational & Skill Enhancement Course

RP/OJT: Research Project / On Job Training

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Semester V

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MCCSCT403 DevOps

B. Sc (Information Technology)		Semester – V		
Course Name: DevOps		Course Code: MCCSCT403		
Vertical:		Major		
Periods per week (1 Period is 60 minutes)		3		
Practical per week (1 Period is 60 minutes)		2		
Credits		4		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Understand the fundamentals of DevOps, including its origins, practices, culture, and its role in accelerating agile development and overcoming technical challenges in the software industry.
2. Explore case studies and best practices in the build, test, deploy, and manage phases of DevOps, focusing on topics such as API management, agile testing, continuous delivery, and agile operations.
3. Learn practical approaches for adopting DevOps within an organization and developing a business case for a DevOps transformation, including defining customer segments, value propositions, revenue streams, and cost structures.
4. Understand the integration of DevOps with enterprise architecture, information security, IT service management, and lean startup methodologies, and explore real-world ROI measurement challenges and strategies in a DevOps context.
5. Explore advanced topics in scaling DevOps for the enterprise, including developing a culture of innovation, continuous improvement, and collaboration, standardizing tools and processes, addressing security considerations, and managing DevOps in outsourcing scenarios.

Module	Name	Lectures
I	DevOps and Agile: Driving Innovation in the Software Lifecycle	9
II	End-to-End API and Software Delivery Lifecycle Management	9
III	Strategic Adoption of DevOps for Business Transformation	9
IV	DevOps: Driving Innovation, Security, and ROI in Modern Enterprises	9
V	Scaling and Leading DevOps Adoption in Enterprises	9
	Total	45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>DevOps and Agile: Driving Innovation in the Software Lifecycle:</u> A. DevOps: An Overview: Origins, Roots, Practices, Culture. B. DevOps in the Ascendancy: Accelerating Agile Practices in Today's Software Factory, Embracing DevOps in the Application Economy, DevOps as a Critical Requirement, Banking on DevOps Practices, DevOps: A Key Component of Business Agility, DevOps: A Practice for Champions. C. IT Impasse: A World of 'Wicked' Business Problems, The Emergence of Agile Development, Modern Application Architectures, Ending the Technical Impasse. D. DevOps Foundations: What Characterizes DevOps Culture?, Lean Thinking to Reduce Waste, DevOps Metrics.	A) R, U B) R, U C) R, U, AN D) U, A, AN
II	<u>End-to-End API and Software Delivery Lifecycle Management:</u> A. Build: Case Study: IceMobile, From Little API Acorns Big Things Grow, API Management: Stakeholders and Requirements, APIs Are Products, Managing the API Lifecycle, API Management: Essential Integrations, Taking a Strategic Approach, Building an API Future, Faster. B. Test: Case Study: AutoTrader.com, Testing Times, Agile Testing Trifecta. C. Deploy: Case Study: Citrix, Obstacles to Continuous Delivery, Continuous Delivery Maturity, Accelerating Maturity: Three Ways, Essential Toolchain Integrations, Release Automation: Capability Checklist, Recommendations and Action Plan. D. Manage: Case Study: ANZ Bank, More Change, More Complexity, New IT Operations Imperatives, Toward Agile Operations.	A) A, EV, CR B) A, EV, CR C) A, EV, CR D) A, EV, CR
III	<u>Strategic Adoption of DevOps for Business Transformation:</u> A. Adopting DevOps: Developing the Playbook. B. Developing a Business Case for a DevOps Transformation: Developing The Business Case, Completing The Business Model Canvas, Customer Segments, Value Propositions, Channels, Customer Relationships, Revenue Streams, Key Resources, Key Activities, Key Partnerships, Cost Structures.	A) EV, CR B) A, CR C) A, CR

	C. DevOps Plays for Optimizing the Delivery Pipeline: DevOps as an Optimization Exercise, Core Themes, The DevOps Plays, Specializing Core Plays.	
IV	<u>DevOps: Driving Innovation, Security, and ROI in Modern Enterprises:</u> A. Practical DevOps: DevOps and Enterprise Architecture, DevOps and Information Security, DevOps and IT Service Management, DevOps and Lean Startup. B. DevOps and Real World ROI: Quality Is Everyone's Responsibility, Making a Case for DevOps ROI, The Challenge of Measuring DevOps Success, A Real-World Model for DevOps ROI, Measuring the Impact of DevOps Automation. C. DevOps Finetuning: Cultural Recalibration, DevOps and the Internet of Things, DevOps for the Public Sector. D. DevOps Plays for Driving Innovation: Optimize to Innovate, The Uber Syndrome, Innovation and the Role of Technology, Core Themes, Play: Build a DevOps Platform, Play: Deliver Microservices Architectures, Play: Develop an API Economy, Play: Organizing for Innovation.	A) U, AN B) EV, CR C) EV, CR D) AN, CR
V	<u>Scaling and Leading DevOps Adoption in Enterprises:</u> A. Scaling DevOps for the Enterprise: Core Themes, Play: DevOps Center of Competency, Play: Developing Culture of Innovation at Scale, Play: Developing a Culture of Continuous Improvement, Play: Team Models for DevOps, Play: Standardization of Tools and Processes, Play: Security Considerations for DevOps, Play: DevOps and Outsourcing. B. Leading DevOps Adoption in the Enterprise: Play: DevOps as a Transformation Exercise, Play: Developing a Culture of Collaboration and Trust, Play: DevOps Thinking for the Line of Business, Play: Starting with Pilot Projects, Play: Rearing Unicorns on an Aircraft Carrier.	A) EV, CR B) EV, CR

List of Practical:		
SR No	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	Introduction to DevOps Tools and Practices:	R, U
a.	Research and document the importance of DevOps in modern software development.	
b.	Install and explore one DevOps tool (e.g., Jenkins, Docker, or Ansible).	
c.	Create a diagram showing how CI/CD, monitoring, and Infrastructure as Code work together	
2.	Building a REST API with Flask or FastAPI	A, CR
a.	Create a simple REST API using Flask or FastAPI with CRUD operations for a "To-Do List" application.	
b.	Test the API using Postman or curl.	
3.	Git Repository Management:	A, CR
a.	Create a Git repository, initialize it, and add a Python project.	
b.	Demonstrate branching by creating a new feature branch, adding code, and merging it back into the main branch using pull requests.	
c.	Simulate a conflict and resolve it.	
4.	Creating and Managing a Virtual Machine:	AN, EV
a.	Set up a virtual machine on a cloud provider or locally (e.g., using VMWare or Azure).	
b.	Configure networking concepts like virtual networks (Vnet), IP addresses, and ports.	
c.	Deploy the Flask/FastAPI app manually on the VM.	
5.	Dockerize a Flask/FastAPI Application:	R, A
a.	Write a Dockerfile for your Flask/FastAPI application and build a Docker image.	
b.	Run the container and test the application.	
c.	Push the Docker image to Docker Hub.	
6.	Docker Compose for Multi-Container Applications:	A, CR
a.	Create a docker-compose.yml file for a multi-container setup (e.g., Flask/FastAPI app + PostgreSQL database).	
b.	Run the application using Docker Compose and test the communication between services.	
7.	Setting Up CI/CD with Jenkins:	AN, EV
a.	Install Jenkins on your virtual machine or local environment.	
b.	Create a pipeline job to automate the build, test, and deployment of your Flask/FastAPI app.	

c.	Integrate Jenkins with Git for automatic build triggers on code commits.	
8.	Ansible Playbooks for Automation:	AN, EV
a.	Write an Ansible playbook to set up and deploy a Python application on multiple servers.	
b.	Include tasks for installing dependencies, configuring the application, and starting the service.	
9.	Monitoring and Observability with Prometheus and Grafana:	EV, AN
a.	Set up Prometheus to collect metrics from a Python application.	
b.	Use Grafana to create dashboards visualizing metrics like CPU usage, memory consumption, and API request rates.	
10.	Deploy a 3-Tier Application:	
a.	Develop a simple 3-tier application with a front-end (e.g., React), back-end (Flask/FastAPI), and database (PostgreSQL).	CR, EV
b.	Dockerize each component and use Docker Compose for orchestration.	
c.	Deploy the application on a virtual machine or local setup and test its functionality.	

Learning Outcomes: After completion of the course students are supposed to be able to:

1. Develop a foundational understanding of DevOps principles, lean thinking, and metrics to drive efficiency and effectiveness in software development and deployment processes.
2. Apply the knowledge gained to develop strategies and techniques for efficient and effective build, test, deploy, and management processes in a DevOps environment, leveraging API management, continuous delivery practices, and agile operations principles.
3. Develop a comprehensive playbook for DevOps adoption, including a business case for transformation, leveraging core DevOps plays, and optimizing the delivery pipeline for improved efficiency and productivity.
4. Implement practical DevOps strategies for enterprise architecture, information security, IT service management, and lean startup initiatives, while effectively measuring the impact and return on investment of DevOps automation.
5. Lead and drive DevOps adoption at the enterprise level by implementing core themes and plays for scaling DevOps, fostering a culture of innovation and collaboration, standardizing processes and tools, and addressing security considerations.

Books and References:

1. Ravichandran, A., Taylor, K., & Waterhouse, P. (2016). DevOps for digital leaders. Apress.
2. Sharma, S. (2017). The DevOps adoption playbook. Wiley & IBM Press.
3. Kim, G., Debois, P., Willis, J., & Humble, J. (2013). The Phoenix project: A novel about IT, DevOps, and helping your business win (1st ed.). IT Revolution Press.
4. Forsgren, N., Humble, J., & Kim, G. (2018). Accelerate: The science of lean software and DevOps: Building and scaling high performing technology organizations (1st ed.). IT Revolution Press.
5. Debois, P., & Shafer, A. C. (2016). The DevOps handbook: How to create world-class agility, reliability, & security in technology organizations (1st ed.). O'Reilly Media.
6. Edwards, D., & Willis, J. (2016). DevOps: A software architect's perspective (1st ed.). Addison-Wesley Professional.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	10	10	20	20	20	20	100%

MCCSCT309 ASP.NET Core

B. Sc (Information Technology)		Semester – V		
Course Name: ASP.NET Core		Course Code: MCCSCT309		
Vertical:		Major		
Periods per week (1 Period is 60 minutes)		03		
Practical per week (1 Period is 60 minutes)		02		
Credits		04		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Understand the core components and architecture of ASP.NET Core, including the application and utility frameworks, and the underlying ASP.NET Core platform.
2. Develop a real-world application using ASP.NET Core, focusing on key functionalities such as data management, navigation, cart functionality, and administration features.
3. Explore security and deployment aspects of ASP.NET Core, including implementing basic authorization policies, creating custom middleware, understanding URL routing, and utilizing platform features like configuration, logging, and exception handling.
4. Gain proficiency in working with data in ASP.NET Core, including caching data and responses, utilizing Entity Framework Core for data access, creating RESTful web services, and implementing advanced features like related data handling and content formatting.
5. Understand the concepts and techniques related to using controllers with views, Razor pages, view components, and tag helpers in ASP.NET Core, including modifying views, utilizing layout and partial views, and creating custom tag helpers.

Module	Name	Lectures
I	Getting Started with ASP.NET Core: Frameworks, Development, and Testing	9
II	Building a Blazor eCommerce Application: From Data Management to Cart and Administration	9
III	Securing, Configuring, and Deploying ASP.NET Core Applications with Advanced Routing and Platform Features	9
IV	Building and Optimizing RESTful Web Services with Data Handling and Caching in ASP.NET Core	9
V	Mastering Views, Razor Pages, and Tag Helpers in ASP.NET Core	9
		45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>Getting Started with ASP.NET Core: Frameworks, Development, and Testing</u> A) Putting ASP.NET Core in Context: Understanding the Application Frameworks, Understanding the Utility Frameworks, Understanding the ASP.NET Core Platform B) Introduction: Choosing a Code Editor, Creating an ASP.NET Core Project, Running the ASP.NET Core Application. C) First ASP.NET Core Application: Setting the Scene, Creating the Project. D) Using the Development Tools: Creating ASP.NET Core Projects, Adding Code and Content to Projects, Building and Running Projects, Managing Packages, Debugging Projects. E) Testing ASP.NET Core Applications: Creating a Unit Test Project, Writing and Running Unit Tests.	A) U,R,A B) U, R, A, AN, CR C) U,R,A,AN D) U, R, A, AN, CR E) U, R, A, AN
II	<u>Building a Blazor eCommerce Application: From Data Management to Cart and Administration</u> A) A Real Application: Creating the Projects: Adding Data to the Application, Displaying a List of Products, Adding Pagination, Styling the Content. B) Navigation and Cart: Adding Navigation Controls, Building the Shopping Cart.	A) U, R, AN, A B) U, R, AN, A

	C) Completing the Cart: Refining the Cart Model with a Service, Completing the Cart Functionality, Submitting Orders. D) Administration: Preparing Blazor Server, Managing Orders, Adding Catalog Management.	C)R, U, A, AN D)U, R, AN, A
III	<u>Securing, Configuring, and Deploying ASP.NET Core Applications with Advanced Routing and Platform Features</u> A) Security and Deployment: Creating the Identity Database, Adding a Conventional Administration Feature, Applying a Basic Authorization Policy, Creating the Account Controller and Views, Testing the Security Policy, Preparing ASP.NET Core for Deployment B) Understanding the ASP.NET Core Platform: Understanding the ASP.NET Core Platform, Understanding the ASP.NET Core Project, Creating Custom Middleware, Configuring Middleware. C) URL Routing: Introduction, Managing URL Matching, Advanced Routing Features. D) Advanced Routing Features: Introduction, Understanding Service Location and Tight Coupling, Using Dependency Injection, Using Service Lifecycles, Other Dependency Injection Features. E) Platform Features: Using the Configuration Service, Using the Logging Service, Using Static Content and Client-Side Packages, Using Cookies, Using Sessions, Working with HTTPS Connections, Handling Exceptions and Errors, Filtering Requests Using the Host Header.	A) R, U, A, AN B) R, U, A, CR C) U, R, AN, A D)R, U, AN, A E)R, U, AN, A
IV	<u>Building and Optimizing RESTful Web Services with Data Handling and Caching in ASP.NET Core</u> A) Working with Data: Caching Data, Caching Responses, Using Entity Framework Core. B) Creating the Example Project: Creating the Project, Adding a Data Model, Adding the CSS Framework, Configuring the Request Pipeline, Running the Example Application. C) Creating RESTful Web Services: Understanding RESTful Web Services, Creating a Web Service Using the Minimal API, Creating a Web Service Using a Controller, Improving the Web Service. D) Advanced Web Service Features: Dealing with Related Data, Supporting the HTTP PATCH Method, Understanding Content Formatting, Documenting and Exploring Web Services.	A) R, U, AN, A, CR B) R, U, AN, A C) R, U, AN, A D) R, U, AN, A
V	<u>Mastering Views, Razor Pages, and Tag Helpers in ASP.NET Core</u> A) Using Controllers with Views: Introduction to views, Modifying a Razor View, Understanding the Razor	

	Syntax, Using the View Bag, Using Temp Data, Working with Layouts, Using Partial Views, Understanding Content-Encoding. B) Using Razor Pages: Understanding Razor Pages, Understanding Razor Pages Routing, Understanding the Page Model Class, Understanding the Razor Page View. C) View Components: Using View Components, Creating and Using a View Component, Understanding View Component Results, Getting Context Data, Creating View Components Classes. D) Tag Helpers: Creating a Tag Helper, Advanced Tag Helper Features, Using Tag Helper Components.	A) R, U, A, AN B) R, U, AN, A, CR C) R, U, AN, A, CR D) R, U, AN, A
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List of Practical:		
Sr. No	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	Creating simple applications and testing applications.	U, A, AN, CR
2.	Creating Testing projects	U, A, AN, CR
3.	Creating Applications with data (CRUD).	U, A, AN
4.	a) Creating application using cart.	U, A, AN, CR
5.	Implementing Security to application.	U, A, AN, CR
6.	Working with URL routing and dependency injections.	U, A, AN, CR
7.	Creating Application with caching	U, R, A, AN, CR
8.	Creating RESTful services	U, R, A, AN, CR
9.	Creating applications with controllers and views	U, A, AN, CR

10.	Creating applications using Razor pages and Helper tags.	U, A, AN, CR

Learning Outcomes: After completion of Course, the learners will be able to:

1. Create and run ASP.NET Core applications using a chosen code editor, demonstrating proficiency in project creation, management, package handling, and debugging techniques.
2. Build a functional ASP.NET Core application that includes features like data display, pagination, shopping cart management, order submission, and administration capabilities.
3. Secure and deploy ASP.NET Core applications by configuring identity databases, applying authorization policies, implementing URL routing, utilizing platform features effectively, and handling exceptions and errors.
4. Develop ASP.NET Core applications that incorporate data caching, Entity Framework Core for data access, RESTful web services with proper content formatting, and advanced features such as handling related data and supporting HTTP PATCH method.
5. Develop ASP.NET Core applications that leverage controllers with views, utilize Razor pages for streamlined development, implement reusable view components, and create custom tag helpers for enhanced functionality and user experience

Books and References:

1. Freeman, A. (2022). Pro ASP.NET Core 6 (9th ed.). Apress.
2. Nagel, C. (2021). Professional C# and .NET (2021 ed.). Wiley.
3. Griffiths, I. (2022). Programming C# 10 (2022 ed.). O'Reilly.
4. Price, M. J. (2022). C# 10 and .NET 6 (6th ed.). Packt.
5. Kanjilal, J. (2019). Mastering C# 8 (2019 ed.). BPB.
6. Skeet, J. (2021). C# in Depth (4th ed.). Manning Publications.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	15	10	15	100%

MCCSCT310 Big Data and NOSQL

B. Sc (Information Technology)		Semester – V		
Course Name: Big Data and NOSQL		Course Code: MCCSCT310		
Vertical:		Major		
Periods per week (1 Period is 60 minutes)		3		
Practical per week (1 Period is 60 minutes)		2		
Credits		4		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. To understand the Big Data and its Use cases.
2. To compare and contrast RDBMS with different NoSQL databases.
3. To understand the architecture of MongoDB.
4. To learn how to run queries in MongoDB in order to store, manipulate, and retrieve data on it.
5. To learn jQuery and its features and understand how to work with data stored in JSON, and creating own JSON strings.

Module	Name	Lectures
I	Big Data and NoSQL: Unlocking the Future of Data Management	9
II	MongoDB Essentials: Data Modeling, Querying, and Advanced System Architecture	9
III	Mastering MongoDB: Storage, Use Cases, Limitations, and Best Practices	9
IV	High-Speed Data and Interactive UIs: Exploring SSDs, In-Memory Databases, and jQuery	9
V	Understanding JSON: A Complete Guide to Syntax, Data Types, and Applications.	9
	Total	45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Details	Level of Knowledge
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		Applicable as per Blooms Taxonomy
I	<u>Big Data and NoSQL: Unlocking the Future of Data Management</u> A. Big Data: Getting Started, Big Data, Facts About Big Data, Big Data Sources, Three Vs of Big Data, Volume, Variety, Velocity, Usage of Big Data, Visibility, Discover and Analyze Information, Segmentation and Customizations, Aiding Decision Making, Innovation, Big Data Challenges, Policies and Procedures, Access to Data, Technology and Techniques, Legacy Systems and Big Data, Structure of Big Data, Data Storage, Data Processing, Big Data Technologies B. NoSQL: SQL, NoSQL, Definition, A Brief History of NoSQL, ACID vs. BASE, CAP Theorem (Brewer's Theorem), The BASE, NoSQL Advantages and Disadvantages, Advantages of NoSQL, Disadvantages of NoSQL, SQL vs. NoSQL Databases, Categories of NoSQL Databases C. Introducing MongoDB: History, MongoDB Design Philosophy, Speed, Scalability, and Agility, Non-Relational Approach, JSON-Based Document Store, Performance vs. Features, Running the Database Anywhere, SQL Comparison	A) R, U, AN, A B) R, U, AN, A C) R, U, AN, A
II	<u>MongoDB Essentials: Data Modeling, Querying, and Advanced System Architecture</u> A. The MongoDB Data Model: The Data Model, JSON and BSON, The Identifier (_id), Capped Collection, Polymorphic Schemas, Object-Oriented Programming, Schema Evolution B. Using MongoDB Shell: Basic Querying, Create and Insert, Explicitly Creating Collections, Inserting Documents Using Loop, Inserting by Explicitly Specifying _id, Update, Delete, Read, Using Indexes, Stepping Beyond the Basics, Using Conditional Operators, Regular Expressions, MapReduce, aggregate(), Designing an Application's Data Model, Relational Data Modeling and Normalization, MongoDB Document Data Model Approach C. MongoDB Architecture: Core Processes, mongod, mongo, mongos, MongoDB Tools, Standalone Deployment, Replication, Master/Slave Replication, Replica Set, Implementing Advanced Clustering with Replica Sets, Sharding, Sharding Components, Data Distribution Process, Data Balancing Process, Operations, Implementing Sharding, Controlling Collection Distribution (Tag-Based Sharding), Points	A) R,U,AN B) R,U,AN,CR,A,EV C) R,U,AN,A

	to Remember When Importing Data in a Sharded Environment, Monitoring for Sharding, Monitoring the Config Servers, Production Cluster Architecture, Scenario 1, Scenario 2, Scenario 3, Scenario 4	
III	<u>Mastering MongoDB: Storage, Use Cases, Limitations, and Best Practices</u> A. MongoDB Storage Engine: Data Storage Engine, Data File (Relevant for MMAPv1), Namespace (.ns File), Data File (Relevant for WiredTiger), Reads and Writes, How Data Is Written Using Journaling, GridFS – The MongoDB File System, The Rationale of GridFS, GridFSunder the Hood, Using GridFS, Indexing, Types of Indexes, Behaviors and Limitations B. MongoDB Use Cases: Use Case 1 -Performance Monitoring, Schema Design, Operations, Sharding, Managing the Data, Use Case 2 – Social Networking, Schema Design, Operations, Sharding C. MongoDB Limitations: MongoDB Space Is Too Large (Applicable for MMAPv1), Memory Issues (Applicable for Storage Engine MMAPv1), 32-bit vs. 64-bit, BSON Documents, Namespaces Limits, Indexes Limit, Capped Collections Limit - Maximum Number of Documents in a Capped Collection, Sharding Limitations, Shard Early to Avoid Any Issues, Shard Key Can't Be Updated, Shard Collection Limit, Select the Correct Shard Key, Security Limitations, No Authentication by Default, Traffic to and from MongoDB Isn't Encrypted, Write and Read Limitations, Case-Sensitive Queries, Type-Sensitive Fields, No JOIN, Transactions, MongoDB Not Applicable Range D. MongoDB Best Practices: Deployment, Hardware Suggestions from the MongoDB Site, Few Points to be Noted, Coding, Application Response Time Optimization, Data Safety, Administration, Replication Lag, Sharding, Monitoring	A) R,U,AN B) R,U,AN C) R,U,AN D) R,U,AN
IV	<u>High-Speed Data and Interactive UIs: Exploring SSDs, In-Memory Databases, and jQuery</u> A. The End of Disk? SSD and In-Memory Databases: The End of Disk?, Solid State Disk, The Economics of Disk, SSD-Enabled Databases, In-Memory Databases, TimesTen, Redis, SAP HANA, VoltDB, Oracle 9c “in-Memory Database, Berkeley Analytics Data Stack and Spark, Spark Architecture B. jQuery: Introduction, Traversing the DOM, DOM Manipulation with jQuery, Events, Ajax with jQuery, jQuery Plug-ins, jQuery Image Slider	A) R,U,AN B) R,U,AN,CR,A,EV

V	<u>Understanding JSON: A Complete Guide to Syntax, Data Types, and Applications</u> A. JSON: Introduction, JSON Grammar, JSON Values, JSON Tokens, Syntax, JSON vs XML, Data Types, Objects, Arrays, Creating JSON, JSON Object, Parsing JSON, Persisting JSON, Data Interchange, JSON PHP, JSON HTML, JSONP	A) R,U,AN,CR,A,EV

List of Practical:		
Sr. No.	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	MongoDB Basics	R, U, A, CR
a	Write a MongoDB query to create and drop database.	
b	Write a MongoDB query to create, display and drop collection	
c	Write a MongoDB query to insert, query, update and delete a document.	
2.	Simple Queries with MongoDB	R, U, A, CR
3.	Implementing Aggregation	R, U, A, CR
a	Write a MongoDB query to use sum, avg, min and max expression.	
b	Write a MongoDB query to use push and addToSet expression.	
c	Write a MongoDB query to use first and last expression.	
4.	Replication, Backup and Restore	R, U, A, CR
a	Write a MongoDB query to create Replica of existing database.	
b	Write a MongoDB query to create a backup of existing database.	

c	Write a MongoDB query to restore database from the backup.	
5.	Java and MongoDB	R, U, A, CR
a	Connecting Java with MongoDB and inserting, retrieving, updating and deleting.	
6.	PHP and MongoDB	R, U, A, CR
a	Connecting PHP with MongoDB and inserting, retrieving, updating and deleting.	
7.	Python and MongoDB	R, U, A, CR
a	Connecting Python with MongoDB and inserting, retrieving, updating and deleting	
8.	Programs on Basic jQuery	R, U, A, CR
a	jQuery Basic, jQuery Events	
b	jQuery Selectors, jQuery Hide and Show effects	
c	jQuery fading effects, jQuery Sliding effects	
9.	jQuery Advanced	R, U, A, CR
a	jQuery Animation effects, jQuery Chaining	
b	jQuery Callback, jQuery Get and Set Contents	
c	jQuery Insert Content, jQuery Remove Elements and Attribute	
10.	JSON and MongoDB	R, U, A, CR
a	Creating, Parsing and Persisting JSON.	
b	Export MongoDB to JSON.	
c	Write a MongoDB query to delete JSON object from MongoDB	

Learning Outcomes: After completion of the course students are supposed to be able to:

1. **Analyze** the differences between Big Data and NoSQL databases, and **evaluate** how Big Data technologies and NoSQL databases like MongoDB address modern data management challenges.
2. **Apply** MongoDB's data modeling techniques, including schema design and querying, to build efficient and scalable data structures for a real-world application. Implement the Replication and Sharding using MongoDB.
3. **Evaluate** the advantages and limitations of MongoDB's storage engines, replication, and sharding strategies, and **design** a robust MongoDB architecture for a high-performance application.
4. **Analyze** the impact of SSD and in-memory databases on application performance, and **apply** jQuery for creating dynamic and interactive user interfaces in web development.
5. **Create** and **parse** JSON data in various formats (e.g., JSONP, JSON in PHP/HTML) and **evaluate** its advantages over XML for data interchange in web applications.

Books and References:

1. Gupta, S. (2015). Practical MongoDB: Architecting, Developing, and Administering MongoDB. Apress.
2. Franklin, J., & Ferguson, R. (2017). Beginning jQuery: From the Basics of jQuery to Writing your Own Plug-ins (2nd ed.). Apress.
3. Harrison, G. (2015). Next Generation Databases: NoSQL and Big Data. Apress.
4. Smith, B. (2015). Beginning JSON (1st ed.). Apress.
5. Banker, K. (2016). MongoDB: The Definitive Guide (2nd ed.). O'Reilly Media.
6. Nadel, B. (2011). Learning jQuery (1st ed.). Packt Publishing.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	10	10	20	100%

MCCSCT404 Enterprise Java

B. Sc (Information Technology)		Semester – V		
Course Name: Enterprise Java		Course Code: MCCSCT404		
Vertical:		Major		
Periods per week (1 Period is 60 minutes)		3		
Practical per week (1 Period is 60 minutes)		2		
Credits		4		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Gain a comprehensive understanding of Java Enterprise Edition (Java EE) and its role in developing enterprise applications, including knowledge of Java EE technologies, architecture, and server/container concepts.
2. Learn advanced concepts in Java Servlets, including request dispatching, working with cookies and sessions, and handling file uploads and downloads, to enhance the functionality and interactivity of web applications.
3. Acquire a thorough understanding of Java Server Pages (JSP) and its advantages over servlets, including knowledge of JSP lifecycle, action elements, implicit objects, and the usage of Java Server Pages Standard Tag Libraries (JSTL).
4. Explore Enterprise JavaBeans (EJB) and its role in enterprise application development, including understanding the architecture, types of session beans, message-driven beans, interceptors, and the Java Naming and Directory Interface (JNDI).
5. Understand the concept of persistence in Java and Object/Relational Mapping (ORM), and gain proficiency in using Java Persistence API (JPA) and Hibernate frameworks for database interaction and management.

Module	Name	Lectures
I	Building Enterprise Applications with Java EE: An Introduction to Servlets, Architecture, and Database Integration	9
II	Advanced Java Servlets: Request Dispatchers, Sessions, Cookies, File Handling, and Non-Blocking I/O	9
III	Mastering Java Server Pages (JSP): Basics, Features, and Best Practices for Dynamic Web Development	9
IV	Exploring Enterprise JavaBeans (EJB): Architecture, Session Beans, Message-Driven Beans, and JNDI Integration	9
V	Mastering Persistence in Java: JPA and Hibernate for Object-Relational Mapping (ORM) and Web Application Development	9
	Total	45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Details	Level of Knowledge Applicable as per Blooms Taxonomy
I	Building Enterprise Applications with Java EE: An Introduction to Servlets, Architecture, and Database Integration: A. Understanding Java EE: What is an Enterprise Application? What is java enterprise edition? Java EE Technologies, Java EE evolution, Glassfish server B. Java EE Architecture, Server and Containers: Types of System Architecture, Java EE Server, Java EE Containers. C. Introduction to Java Servlets: The Need for Dynamic Content, Java Servlet Technology, Why Servlets? What can Servlets do? D. Servlet API and Lifecycle: Java Servlet API, The Servlet Skeleton, The Servlet Life Cycle, A Simple Welcome Servlet E. Working with Servlets: Getting Started, Using Annotations Instead of Deployment Descriptor. F. Working with Databases: What Is JDBC? JDBC Architecture, Accessing Database, The Servlet GUI and Database Example.	A) U, R B) U, R C) U, EV D) U, R, EV E) U, AN F) U, EV
II	Advanced Java Servlets: Request Dispatchers, Sessions, Cookies, File Handling, and Non-Blocking I/O: A. Request Dispatcher: Requestdispatcher Interface, Methods of Requestdispatcher, Requestdispatcher Application. B. COOKIES: Kinds of Cookies, Where Cookies Are Used? Creating Cookies Using Servlet, Dynamically Changing the Colors of A Page C. SESSION: What Are Sessions? Lifecycle of Http Session, Session Tracking With Servlet API, A Servlet Session Example D. Working with Files: Uploading Files, Creating an Upload File Application, Downloading Files, Creating a Download File Application. E. Working with Non-Blocking I/O: Creating a Non-Blocking Read Application, Creating The Web Application, Creating Java Class, Creating Servlets, Retrieving The File, Creating index.jsp	A) U, A, AN B) U, EV, CR C) U, R, EV D) R, CR E) CR, EV
III	Mastering Java Server Pages (JSP): Basics, Features, and Best Practices for Dynamic Web Development:	A) U, R

	A. Introduction To Java Server Pages: Why use Java Server Pages? Disadvantages Of JSP, JSP v\s Servlets, Life Cycle of a JSP Page, How does a JSP function? How does JSP execute? About Java Server Pages B. Getting Started With Java Server Pages: Comments, JSP Document, JSP Elements, JSP GUI Example. C. Action Elements: Including other Files, Forwarding JSP Page to Another Page, Passing Parameters for other Actions, Loading a Javabean. D. Implicit Objects, Scope and El Expressions: Implicit Objects, Character Quoting Conventions, Unified Expression Language [Unified El], Expression Language. E. Java Server Pages Standard Tag Libraries: What is wrong in using JSP Scriptlet Tags? How JSTL Fixes JSP Scriptlet's Shortcomings? Disadvantages Of JSTL, Tag Libraries.	B) U, EV C) U, R, EV D) U, AN E) U, EV
IV	Exploring Enterprise JavaBeans (EJB): Architecture, Session Beans, Message-Driven Beans, and JNDI Integration: A. Introduction To Enterprise Javabeans: Enterprise Bean Architecture, Benefits of Enterprise Bean, Types of Enterprise Bean, Accessing Enterprise Beans, Enterprise Bean Application, Packaging Enterprise Beans B. Working with Session Beans: When to use Session Beans? Types of Session Beans, Remote and Local Interfaces, Accessing Interfaces, Lifecycle of Enterprise Beans, Packaging Enterprise Beans, Example of Stateful Session Bean, Example of Stateless Session Bean, Example of Singleton Session Beans. C. Working with Message Driven Beans: Lifecycle of a Message Driven Bean, Uses of Message Driven Beans, The Message Driven Beans Example. D. Interceptors: Request and Interceptor, Defining An Interceptor, AroundInvoke Method, Applying Interceptor, Adding An Interceptor To An Enterprise Bean, Build and Run the Web Application. E. Java Naming and Directory Interface: What is Naming Service? What is Directory Service? What is Java Naming and Directory interface? Basic Lookup, JNDI Namespace in Java EE, Resources and JNDI, Datasource Resource Definition in Java EE.	A) U, EV B) A, R, CR C) U, R, CR D) AN, CR, R E) U, R, CR
V	Mastering Persistence in Java: JPA and Hibernate for Object-Relational Mapping (ORM) and Web Application Development: A. Persistence, Object/Relational Mapping And JPA: What is Persistence? Persistence in Java, Current Persistence Standards in Java, Why another Persistence Standards? Object/Relational Mapping,	A) U, R

	B. Introduction to Java Persistence API: The Java Persistence API, JPA, ORM, Database and the Application, Architecture of JPA, How JPA Works? JPA Specifications. C. Writing JPA Application: Application Requirement Specifications, Software Requirements, The Application Development Approach, Creating Database and Tables in Mysql, creating a Web Application, Adding the Required Library Files, creating a Javabeen Class, Creating Persistence Unit [Persistence.Xml], Creating JSPS, The JPA Application Structure, Running the JPA Application. D. Introduction to Hibernate: What is Hibernate? Why Hibernate? Hibernate, Database and The Application, Components of Hibernate, Architecture of Hibernate, How Hibernate Works? E. Writing Hibernate Application: Application Requirement Specifications, Software Requirements, The Application Development Approach, Creating Database and Tables in Mysql, creating a Web Application, Adding the Required Library Files, creating a Javabeen Class, Creating Hibernate Configuration File, Adding a Mapping Class, Creating JSPS, Running The Hibernate Application.	B) R, U, EV C) U, AN D) U, R, AN E)U, R, EV
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List of Practical		
Sr. No.	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	Implement the following Simple Servlet applications.	CR, AN
a.	Create a simple calculator application using servlet.	
b.	Create a servlet for a login page. If the username and password are correct then it says message “Hello <username>” else a message “login failed”	
c.	Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.	
2.	Implement the following Servlet applications with Cookies and Sessions.	CR, AN, U
a.	Using Request Dispatcher Interface create a Servlet which will validate the password entered by the user, if the user has entered	

	"Servlet" as password, then he will be forwarded to Welcome Servlet else the user will stay on the index.html page and an error message will be displayed.	
b.	Create a servlet that uses Cookies to store the number of times a user has visited servlet.	
c.	Create a servlet demonstrating the use of session creation and destruction. Also check whether the user has visited this page first time or has visited earlier also using sessions.	
3.	Implement the Servlet IO and File applications.	CR, R, AN
a.	Create a Servlet application to upload and download a file.	
b.	Develop Simple Servlet Question Answer Application using Database.	
c.	Create simple Servlet application to demonstrate Non-Blocking Read Operation.	
4.	Implement the following JSP applications.	CR, AN, U
a.	Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types.	
b.	Develop a simple JSP application to pass values from one page to another with validations. (Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).	
c.	Create a registration and login JSP application to register and authenticate the user based on username and password using JDBC.	
5.	Implement the following JSP JSTL and EL Applications.	CR, AN, R
a.	Create an html page with fields, eno, name, age, desg, salary. Now on submit this data to a JSP page which will update the employee table of database with matching eno.	
b.	Create a JSP page to demonstrate the use of Expression language.	
c.	Create a JSP application to demonstrate the use of JSTL.	
6.	Implement the following EJB Applications.	CR, AN, A
a.	Create a Currency Converter application using EJB.	
b.	Develop a Simple Room Reservation System Application Using EJB.	
c.	Develop simple shopping cart application using EJB [Stateful Session Bean].	
7.	Implement the following EJB applications with different types of Beans.	CR, A

a.	Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.	
b.	Develop simple visitor Statistics application using Message Driven Bean [Stateless Session Bean].	
c.	Develop simple Marks Entry Application to demonstrate accessing Database using EJB.	
8.	Implement the following JPA applications.	CR, A, U
a.	Develop a simple Inventory Application Using JPA.	
b.	Develop a Guestbook Application Using JPA.	
c.	Create simple JPA application to store and retrieve Book details.	
9.	Implement the following JPA applications with ORM and Hibernate.	CR, A, AN
a.	Develop a JPA Application to demonstrate use of ORM associations.	
b.	Develop a Hibernate application to store Feedback of Website Visitor in MySQL Database.	
c.	Develop a Hibernate application to store and retrieve employee details in MySQL Database.	
10.	Implement the following Hibernate applications.	CR, U, A
a.	Develop an application to demonstrate Hibernate One- To -One Mapping Using Annotation.	
b.	Develop Hibernate application to enter and retrieve course details with ORM Mapping.	
c.	Develop a five page web application site using any two or three Java EE Technologies.	

Learning Outcomes:After completion of the course students are supposed to be able to:

1. Develop the ability to design and implement dynamic web applications using Java Servlets and integrate them with databases, leveraging the Java Servlet API and JDBC for efficient data access.
2. Apply the knowledge gained to build robust web applications using Java Servlets, incorporating features such as request dispatching, cookie management, session tracking, and file handling.
3. Develop the ability to design and implement dynamic web pages using JSP, utilizing JSP elements, expressions, and JSTL tags to enhance code readability, modularity, and maintainability.
4. Apply the knowledge gained to develop and deploy enterprise applications using EJB, leveraging session beans, message-driven beans, interceptors, and JNDI for scalable and maintainable solutions.

5. Develop the ability to build database-driven applications using JPA and Hibernate, including designing and implementing entity classes, configuring persistence units, and performing database operations effectively.

Books and References:

1. Shah, S., & Shah, V. (2017). Java EE 7 for beginners (1st ed.). SPD.
2. Moraes, E. (2018). Java EE 8 cookbook: Build reliable applications with the most robust and mature technology for enterprise development (1st ed.). Packt.
3. Roy, U. K. (2015). Advanced Java programming. Oxford Press.
4. Goyal, N. K. (2018). Java EE 8: The complete reference (1st ed.). McGraw-Hill Education.
5. G, R. S. (2019). Mastering Java EE development with WildFly (1st ed.). Packt Publishing.
6. K, R. S. (2018). Spring 5 and Java EE 8: A guide to building microservices (1st ed.). Wiley.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	30	10	10	5	25	100%

MCCSCT306 Internet of Things

B. Sc (Information Technology)		Semester – V		
Course Name: Internet of Things		Course Code: MCCSCT306		
Vertical:		Minor		
Periods per week (1 Period is 60 minutes)		03		
Practical per week (1 period is 60 minutes)		02		
Credits		04		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	-	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	-	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Provide a comprehensive understanding of the Internet of Things (IoT), covering its fundamental concepts, communication protocols, design principles, and real-world applications.
2. Covers IoT architecture, communication models, and protocols like TCP/IP, HTTP, and MQTT for efficient data exchange in IoT networks.
3. Introduces hardware platforms such as Arduino and Raspberry Pi, embedded programming, sensor integration, and cloud connectivity for IoT applications.
4. Explores IoT security challenges, data privacy issues, and ethical concerns related to data ownership, control, and sustainability.
5. Equips students with a strong foundation in IoT concepts, hands-on prototyping, embedded systems programming, security and privacy measures, business and startup opportunities, manufacturing processes, ethical considerations, and industry-relevant skills for career growth.

Module	Name	Lectures
I	The Internet of Things: An Overview and Design Principles	9
II	Prototyping Concepts and Embedded Devices	9
III	Prototyping Physical and Online Components	9
IV	Embedded Code Techniques and Business Models	9
V	Manufacturing Processes and Ethical Consideration	9
		45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus As per SSC framework of NEP	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>The Internet of Things: An Overview and Design Principles</u> A) The Internet of Things: An Overview: The Flavour of the Internet of Things, The “Internet” of “Things”, The Technology of the Internet of Things, Enchanted Objects, Who is Making the Internet of Things? B) Design Principles for Connected Devices: Calm and Ambient Technology, Magic as Metaphor, Privacy, Keeping Secrets, Whose Data Is It Anyway? Web Thinking for Connected Devices, Small Pieces, Loosely Joined, First-Class Citizens on The Internet, Graceful Degradation, and Affordances. C) Internet Principles: Internet Communications: An Overview, IP, TCP, The IP Protocol Suite (TCP/IP), UDP, IP Addresses, DNS, Static IP Address Assignment, Dynamic IP Address Assignment, IPv6, MAC Addresses, TCP and UDP Ports, An Example: HTTP Ports, Other Common Ports, Application Layer Protocols, HTTP, HTTPS: Encrypted HTTP, Other Application Layer Protocols.	A) R, U B) U, A, AN, EV C) R, U, A
II	<u>Prototyping Concepts and Embedded Devices</u> A) Thinking About Prototyping: Sketching, Familiarity, Costs versus Ease of Prototyping, Prototypes and Production, Changing Embedded Platform, Physical Prototypes and Mass Personalisation, climbing into the Cloud, Open Source versus Closed Source, Why Closed? Why Open? Mixing Open and Closed Source, Closed Source for Mass Market Projects, Tapping into the Community. B) Prototyping Embedded Devices: Electronics, Sensors, Actuators, Scaling Up the Electronics, Embedded Computing Basics, Microcontrollers, System-on-Chips, Choosing Your Platform, Arduino, developing on the Arduino, Some Notes on the Hardware, Openness, Raspberry Pi, Cases and Extension Boards, Developing on the Raspberry Pi, Some Notes on the Hardware, Openness.	A) R, U, A, AN, EV B) R, U, A, AN
III	<u>Prototyping Physical and Online Components</u> A) Prototyping the Physical Design: Preparation, Sketch, Iterate, and Explore, Nondigital Methods, Laser Cutting,	A) U, A, EV, CR

	Choosing a Laser Cutter, Software, Hinges and Joints, 3D Printing, Types of 3D Printing, Software, CNC Milling, Repurposing/Recycling. B) Prototyping Online Components: Getting Started with an API, Mashing Up APIs, Scraping, Legalities, writing a New API, Clockodillo, Security, implementing the API, Using Curl to Test, Going Further, Real-Time Reactions, Polling, Comet, Other Protocols, MQ Telemetry Transport, Extensible Messaging and Presence Protocol, Constrained Application Protocol.	B) R, U, A, EV, CR
IV	<u>Embedded Code Techniques and Business Models</u> A) Techniques for Writing Embedded Code: Memory Management, Types of Memory, Making the Most of Your RAM, Performance and Battery Life, Libraries, Debugging. B) Business Models: A Short History of Business Models, Space and Time, From Craft to Mass Production, The Long Tail of the Internet, Learning from History, The Business Model Canvas, Who Is the Business Model For? Models, Make Thing, Sell Thing, Subscriptions, Customization, be a Key Resource, Provide Infrastructure: Sensor Networks, take a Percentage, Funding an Internet of Things Startup, Hobby Projects and Open Source, Venture Capital, Government Funding, Crowdfunding, Lean Startups.	A) U, A, EV B) R, U, AN, EV, CR
V	<u>Manufacturing Processes and Ethical Consideration</u> A) Moving to Manufacture: What Are You Producing? Designing Kits, Designing Printed circuit boards, Software Choices, The Design Process, Manufacturing Printed Circuit Boards, Etching Boards, Milling Boards. Assembly, Testing, Mass-Producing the Case and Other Fixtures, Certification, Costs, Scaling Up Software, Deployment, Correctness and Maintainability, Security, Performance, User Community. B) Ethics: Characterizing the Internet of Things, Privacy, Control, Disrupting Control, Crowdsourcing, Environment, Physical Thing, Electronics, Internet Service, Solutions, The Internet of Things as Part of the Solution, Cautious Optimism, The Open Internet of Things Definition.	A) R, U, A, AN, EV, CR B) U, AN, EV

Internet of Things – Practical		
Sr. No.	List of Practicals	Level of Knowledge Applicable as per Blooms Taxonomy
0	Starting Raspbian OS, Familiarizing with Raspberry Pi Components and interface, Connecting to ethernet, Monitor, USB.	R, U, A
1	Displaying different LED patterns with Raspberry Pi.	R, U, A, CR
2	Displaying Time over 4-Digit 7-Segment Display using Raspberry Pi.	U, A, CR
3	Raspberry Pi Based Oscilloscope	R, U, A, AN, CR
4	Controlling Raspberry Pi with Telegram bot.	U, A, CR
5	Fingerprint Sensor interfacing with Raspberry Pi	R, U, A, CR
6	Raspberry Pi GPS Module Interfacing	A, CR
7	IoT based Web Controlled Home Automation using Raspberry Pi	U, A, AN, CR
8	Visitor Monitoring with Raspberry Pi and Pi Camera	U, A, AN, CR
9	Interfacing Raspberry Pi with RFID	R, U, A, CR
10	Installing Windows 10 IoT Core on Raspberry Pi	U, A

Learning Outcomes: After completion of the Course, the learners will be able to:

1. Understand IoT fundamentals, technologies, design principles, communication protocols, and security to develop efficient and scalable IoT solutions.
2. Design and implement IoT architectures and prototyping techniques using embedded systems, including Arduino, Raspberry Pi, sensors, actuators, gateways, cloud-based systems, and data analytics.
3. Gain proficiency in prototyping physical and online components using digital fabrication techniques, APIs, real-time communication protocols, and security considerations.
4. Develop embedded code with efficient memory management, debugging, and performance optimization while understanding IoT business models, funding, and commercialization strategies.
5. Learn manufacturing processes, PCB design, mass production techniques, and ethical considerations in IoT, including privacy, security, sustainability, and open-source development.

Books & References:

1. McEwen, A., & Cassimally, H. (2014). *Designing the Internet of Things* (1st ed.). Wiley.
2. Kamal, R. (2017). *Internet of Things – Architecture and Design* (1st ed.). McGraw-Hill.
3. Pfister, C. (2018). *Getting Started with the Internet of Things* (6th ed.). O'Reilly.
4. Greengard, S. (2015). *The Internet of Things* (1st ed.). MIT Press.
5. Bahga, A., & Madiseti, V. (2014). *Internet of Things: A Hands-On Approach* (1st ed.). VPT.

6. Margolis, M. (2020). *Arduino Cookbook* (3rd ed.). O'Reilly Media.
7. Monk, S. (2013). *Programming the Raspberry Pi: Getting Started with Python* (2nd ed.). McGraw-Hill Education.
8. Marwedel, P. (2021). *Embedded System Design: Embedded Systems Foundations of Cyber-Physical Systems* (3rd ed.). Springer.
9. Rose, K. (2014). *The Internet of Things: Enabling Technologies, Platforms, and Use Cases* (1st ed.). Wiley.
10. Pethuru, R., Anupama, C. R., & Vijay, M. (2017). *The Internet of Things: Enabling Technologies, Security and Social Implications* (1st ed.). CRC Press.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	15	25	20	25	10	05	100%

MCCSCT311 Artificial Intelligence

B. Sc (Information Technology)		Semester – V		
Course Name: Artificial Intelligence		Course Code: MCCSCT311		
Vertical:		VSEC		
Periods per week (1 Period is 60 minutes)		3		
Practical per week (1 Period is 60 minutes)		2		
Credits		4		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Understand the fundamental concepts and foundations of Artificial Intelligence (AI), including its history, current state, and various subfields within AI.
2. Develop problem-solving skills using search algorithms in AI, including both uninformed and informed search strategies, and understand the role of heuristic functions in guiding search.
3. Explore adversarial search in AI, focusing on games as the domain, and understand concepts such as optimal decision-making, alpha-beta pruning, stochastic games, and partially observable games.
4. Gain knowledge and skills in logical reasoning and knowledge representation using First Order Logic (FOL), including syntax, semantics, and reasoning techniques.
5. Explore advanced topics in AI, including planning methodologies, multi-agent planning, and knowledge representation beyond propositional logic, such as reasoning with categories, events, and default information.

Module	Name	Lectures
I	Introduction and Intelligent Agents	9
II	Problem Solving: Classical and Advanced Search Methods	9
III	AI Agents: Adversarial Search and Logical Reasoning	9
IV	Unification in First-Order Logic and Its Role in Inference	9
V	Planning and Knowledge Representation	9
	Total	45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	Introduction and Intelligent Agents: - A. Introduction: What is Artificial Intelligence? Foundations of AI, history, Nyaya Philosophy (Logic and Categorization), Paninian Grammar (Linguistic Classification), the state of art AI today. B. Intelligent Agents: agents and environment, good behavior, nature of environment, the structure of agents.	A)U, R, AN B) U, R, AN
II	Problem Solving: Classical and Advanced Search Methods: A. Solving Problems by Searching: Problem solving agents, examples problems, searching for solutions, uninformed search, informed search strategies, heuristic functions. B. Beyond Classical Search: local search algorithms, searching with non-deterministic action, searching with partial observations, online search agents and unknown environments.	A) U, AN, EV B) U, R, AN
III	AI Agents: Adversarial Search and Logical Reasoning: A. Adversarial Search: Games, optimal decisions in games, alpha-beta pruning, stochastic games, partially observable games, state-of-the-art game programs. B. Logical Agents: Knowledge base agents, The Wumpus world, logic, propositional logic, propositional theorem proving, effective propositional model checking, agents based on propositional logic.	A) U, R, AN B) U, R, AN
IV	Unification in First-Order Logic and Its Role in Inference: A. First Order Logic: Syntax and semantics, using First Order Logic, Knowledge engineering in First Order Logic. B. Inference in First Order Logic: propositional vs. First Order, unification and lifting, forward and backward chaining, resolution.	A) U, R, AN B) U, R, EV
V	Planning and Knowledge Representation: A. Planning: Definition of Classical Planning, Algorithms for planning as state space search, planning graphs, other classical planning approaches, analysis of planning approaches, Time, Schedules and resources,	A) U, R, AN

	hierarchical planning, Planning and Acting in Nondeterministic Domains, multiagent planning, B. Knowledge Representation: Categories and Objects, events, mental events and objects, reasoning systems for categories, reasoning with default information, Internet shopping world.	B) U, R, AN
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List of Practicals		
The practicals can be implemented in C / C++ / Java/ Python / R /Prolog / LISP or any other language.		
Sr. No.	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1	a Write a program to implement depth first search algorithm.	R, U
	b Write a program to implement breadth first search algorithm.	
2	a Write a program to simulate 4-Queen / N-Queen problem.	A, CR
	b Write a program to solve tower of Hanoi problem.	
3	a Write a program to implement alpha beta search.	A, CR
	b Write a program for Hill climbing problem.	
4	a Write a program to implement A* algorithm.	AN, EV
	b Write a program to implement AO* algorithm.	
5	a Write a program to solve water jug problem.	R, A
	b Design the simulation of tic – tac – toe game using min-max algorithm.	
6	a Write a program to solve Missionaries and Cannibals problem.	A, CR
	b Design an application to simulate number puzzle problem.	
7	a Write a program to shuffle Deck of cards.	AN, EV
	b Solve traveling salesman problem using artificial intelligence technique.	
8	a Solve the block of World problem.	EV, CR
	b Solve constraint satisfaction problem	
9	a Derive the expressions based on Associative law	CR, A

	b	Derive the expressions based on Distributive law	
10	a	Write a program to derive the predicate. (for e.g.: Sachin is batsman, batsman is cricketer) - > Sachin is Cricketer.	R, A
	b	Write a program which contains three predicates: male, female, parent. Make rules for following family relations: father, mother, grandfather, grandmother, brother, sister, uncle, aunt, nephew and niece, cousin. Question: i. Draw Family Tree. ii. Define: Clauses, Facts, predicates and rules with conjunction and disjunction	R, A

Learning Outcomes: After completion of the course students are supposed to be able to:

1. Gain a comprehensive overview of AI, enabling students to explain the core principles, advancements, and applications of AI in contemporary society.
2. Apply different search algorithms and heuristic functions to solve a variety of problems, demonstrating proficiency in designing efficient and effective problem-solving agents.
3. Develop strategies and algorithms to make optimal decisions in adversarial environments, demonstrating proficiency in building game-playing agents that can compete against human players or state-of-the-art game programs.
4. Apply First Order Logic to represent and reason about knowledge in various domains, demonstrating proficiency in knowledge engineering and building intelligent agents based on FOL.
5. Design and implement planning algorithms to solve complex problems, including resource allocation, scheduling, and hierarchical planning. Develop expertise in representing and reasoning about knowledge in diverse domains, including event-driven systems and reasoning with defaults.

Books and References:

1. Russell, S., & Norvig, P. (2015). Artificial intelligence: A modern approach (3rd ed.). Pearson.
2. Khemani, D. (2017). A first course in artificial intelligence (1st ed.). TMH.
3. Matilal, B. K. (1990). Logic, language, and reality: Indian philosophy (2nd ed.)., Delhi: Motilal Banarsidass Publishers Private.
4. Deva, R. (2018). Artificial intelligence: A rational approach (1st ed.). Shroff Publishers.
5. Rich, E., Knight, K., & Nair, S. (2009). Artificial intelligence (3rd ed.). TMH.
6. Bhattacharjee, A. D. (2013). Artificial intelligence & soft computing for beginners (1st ed.). SPD.
7. Nilsson, N. J. (1998). Artificial intelligence: A new synthesis (1st ed.). Morgan Kaufmann Publishers.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	10	10	20	100%

MCCSCT307 Virtual Reality and Augmented Reality

B. Sc (Information Technology)		Semester – V		
Course Name: Virtual Reality and Augmented Reality		Course Code: MCCSCT307		
Vertical:		VSEC		
Periods per week (1 Period is 60 minutes)		03		
Practical per week (1 Period is 60 minutes)		02		
Credits		04		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Objectives of the Course:

1. Understand the fundamental concepts and principles of Virtual Reality (VR), including its history, different realities, immersion, presence, and perceptual models.
2. Explore the perceptual aspects of VR, including perception of space and time, perceptual stability, attention, and action. Understand the hardware challenges and adverse health effects associated with VR.
3. Develop skills in VR content creation, including environmental design, influencing user behavior, and interaction design. Explore human-centered interaction principles and various input devices for VR.
4. Examine the intersection of design and art in digital realities, including optimizing 3D art for VR, understanding computer vision in augmented reality, and exploring cross-platform theories.
5. Explore advanced topics and tools in VR, including open-source frameworks, data visualization in spatial computing, character AI and behaviors, and the healthcare technology ecosystem in VR and AR.

Unit/Module	Name	Lectures
I	Exploring Virtual Reality: Concepts, History, Design Principles, and Perception Models	9
II	Perception, Health, and Design in Virtual Reality: Challenges and Guidelines	9
III	Human-Centered VR Content Creation and Interaction Design	9
IV	Design, Art, and Cross-Platform Integration in Digital Realities	9
V	Innovations in Virtual and Augmented Reality: Tools, AI, and Health Ecosystems	9
		45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>Exploring Virtual Reality: Concepts, History, Design Principles, and Perception Models</u> A) Introduction: What Is Virtual Reality, A History of VR, An Overview of Various Realities, Immersion, Presence, and Reality Trade-Offs, The Basics: Design Guidelines, Objective and Subjective Reality, Perceptual Models and Processes, Perceptual Modalities.	A) U,R,A
II	<u>Perception, Health, and Design in Virtual Reality: Challenges and Guidelines</u> A) Perception of Space and Time, Perceptual Stability, Attention, and Action, Perception: Design Guidelines, Adverse Health Effects, Motion Sickness, Eye Strain, Seizures, and Aftereffects, Hardware Challenges, Latency, Measuring Sickness, Reducing Adverse Effects, Adverse Health Effects: Design Guidelines.	A) U, R, AN, A
III	<u>Human-Centered VR Content Creation and Interaction Design</u> A) Content Creation, Concepts of Content Creation, Environmental Design, Affecting Behavior, Transitioning to VR Content Creation, Content Creation: Design Guidelines, Interaction, Human-Centered Interaction, VR Interaction Concepts, Input Devices, Interaction Patterns and Techniques, Interaction: Design Guidelines.	A) R, U, A, AN
IV	<u>Design, Art, and Cross-Platform Integration in Digital Realities</u> A) Design and Art Across Digital Realities, Designing for Our Senses, Virtual Reality for Art, 3D Art Optimization, Computer Vision That Makes Augmented Reality Possible Works, Virtual Reality and Augmented Reality: Cross-Platform Theory.	A) R, U, AN, A, CR
V	<u>Innovations in Virtual and Augmented Reality: Tools, AI, and Health Ecosystems</u> A) Virtual Reality Toolkit: Open Source Framework for the Community, Data and Machine Learning Visualization Design and Development in Spatial Computing, Character AI and Behaviors, The Virtual and Augmented Reality Health Technology Ecosystem.	A) R, U, A. AN

List of Practical:		
Sr. No	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	Build an environment using primitives like boxes, spheres and cylinders	U, A, AN, CR
2.	Develop a simple architecture of a house	U, A, AN, CR
3.	Develop a simple architecture of a house	U, A, AN
4.	Develop a simple game application using 3D modelling and rendering techniques	U, A, AN, CR
5.	Develop a concept of simple virtual classroom	U, A, AN, CR
6.	Show the simulation of a plant growing using augmented reality	U, A, AN, CR
7.	Show the animation of a fish swimming using augmented reality	U, R, A, AN, CR
8.	Simulate a basket ball court using augmented reality	U, R, A, AN, CR
9.	Show the simulation of a moving car using augmented reality	U, A, AN, CR
10.	Develop an application in any domain of interest e.g., Gaming, Human Pose etc	U, A, AN, CR

Learning Outcomes: After completion of Course, the learners will be able to:

1. Gain knowledge and comprehension of VR technologies and concepts, enabling students to analyze and discuss the design guidelines and trade-offs involved in creating immersive virtual experiences.
2. Apply design guidelines to mitigate adverse health effects in VR experiences, demonstrate an understanding of perceptual considerations in VR design, and propose solutions to reduce motion sickness, eye strain, and other adverse effects.
3. Design and create immersive VR content following design guidelines, demonstrate proficiency in creating interactive experiences in VR, and apply human-centered design principles to enhance user engagement and satisfaction.
4. Analyze and evaluate the design and art aspects of VR and AR experiences, demonstrate proficiency in optimizing 3D art for VR platforms, and propose strategies for creating immersive and cross-platform digital realities.
5. Apply virtual reality toolkits and frameworks to develop immersive experiences, demonstrate proficiency in visualizing data and utilizing machine learning techniques in spatial computing, and analyze the potential of VR/AR in healthcare technology.

Books and References:

1. Jerald, J. (2016). The VR book: Human-centered design for virtual reality (1st ed.). ACM Books.
2. Pangilinan, E., Lukas, S., & Mohan, V. (2019). Creating augmented and virtual realities (1st ed.). O'Reilly Media.
3. Baruah, R. (2020). Virtual reality with VRTK4 (1st ed.). Apress.
4. Moore, M. S. (2019). Augmented reality for developers (1st ed.). O'Reilly Media.
5. Scoble, R., & Israel, S. (2016). The fourth transformation: How augmented reality & artificial intelligence will change everything (1st ed.). Greenleaf Book Group Press.
6. Mealy, P. (2017). Unity 3D virtual reality projects (1st ed.). Packt Publishing.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	15	10	15	100%

MCCSCTPRJ401 Project

B. Sc (Information Technology)	Semester – V		
Course Name: Project	Course Code: MCCSCTPRJ401		
Vertical:	RP / OJT		
Periods per week (1 Period is 60 minutes)	2		
Practical per week (1 Period is 60 minutes)	--		
Credits	2		
Evaluation System	Duration (in Hours)	Total Marks	Minimum Passing Marks
Presentation & Viva	2	50	20

The details are given in Project Policy document.

MCCOJTSCT302 On Job Training

B. Sc (Information Technology)	Semester – V		
Course Name: On Job Training	Course Code: MCCOJTSCT302		
Vertical:	RP / OJT		
Periods per week (1 Period is 60 minutes)	2		
Practical per week (1 Period is 60 minutes)	--		
Credits	2		
Evaluation System	Duration (in Hours)	Total Marks	Minimum Passing Marks
Continuous Internal Assessment	--	50	20

The details are given in OJT Policy document.

Semester VI

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MCCSCT405 Computer Security

B. Sc (Information Technology)		Semester – VI		
Course Name: Computer Security		Course Code: MCCSCT405		
Vertical:		Major		
Periods per week (1 Period is 60 minutes)		3		
Practical per week (1 Period is 60 minutes)		2		
Credits		4		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. To develop an understanding of information assets and understanding security philosophy to strategy to technology to behavior.
2. To learn approaches to secure data via encryption and data residing in storage and network.
3. To design a secure network by configuring devices and firewalls and provide wireless network security.
4. To understand the need for Intrusion Detection and Prevention Systems and how to apply computer security in operating systems, virtual machines and clouds.
5. To design a secure application and importance of physical security.

Module	Name	Lectures
I	Foundations of Information Security: Risk Analysis, Secure Design, and Protection Strategies	9
II	Core Aspects of Cybersecurity: Authentication, Encryption, and Data Protection	9
III	Network Security and Design: Safeguarding Infrastructure and Wireless Networks	9
IV	Advanced Security Mechanisms: IDS, VoIP, and Operating System Protection	9
V	Comprehensive Security Framework: Virtualization, Application, and Physical Protection	9
	Total	45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>Foundations of Information Security: Risk Analysis, Secure Design, and Protection Strategies:</u> A. Information Security Overview: The Importance of Information Protection, The Evolution of Information Security, Justifying Security Investment, Security Methodology, How to Build a Security Program, The Impossible Job, The Weakest Link, Strategy and Tactics, Business Processes vs. Technical Controls. B. Risk Analysis: Threat Definition, Types of Attacks, Risk Analysis. C. Secure Design Principles: The CIA Triad and Other Models, Defense Models, Zones of Trust, Best Practices for Network Defense.	A) R, U B) U, A C) A, AN
II	<u>Core Aspects of Cybersecurity: Authentication, Encryption, and Data Protection</u> A. Authentication and Authorization: Authentication, Authorization B. Encryption: A Brief History of Encryption, Symmetric-Key Cryptography, Public Key Cryptography, Public Key Infrastructure. C. Storage Security: Storage Security Evolution, Modern Storage Security, Risk Remediation, Best Practices. D. Database Security: General Database Security Concepts, Understanding Database Security Layers, Understanding Database-Level Security, Using Application Security, Database Backup and Recovery, Keeping Your Servers Up to Date, Database Auditing and Monitoring.	A) U, EV B) U, EV C) U, EV D) U, EV
III	<u>Network Security and Design: Safeguarding Infrastructure and Wireless Networks</u> A. Secure Network Design: Introduction to Secure Network Design, Performance, Availability, Security. B. Network Device Security: Switch and Router Basics, Network Hardening. C. Firewalls: Overview, The Evolution of Firewalls, Core Firewall Functions, Additional Firewall Capabilities, Firewall Design. D. Wireless Network Security: Radio Frequency Security Basics, Data-Link Layer Wireless Security Features, Flaws, and Threats, Wireless Vulnerabilities and Mitigations, Wireless Network Hardening Practices and Recommendations, Wireless Intrusion Detection and Prevention, Wireless Network Positioning and Secure Gateways.	A) R, EV B) R, EV C) U, EV D) U, AN

IV	<u>Advanced Security Mechanisms: IDS, VoIP, and Operating System Protection:</u> A. Intrusion Detection and Prevention Systems: IDS Concepts, IDS Types and Detection Models, IDS Features, IDS Deployment Considerations, Security Information and Event Management (SIEM). B. Voice over IP (VoIP) and PBX Security: Background, VoIP Components, VoIP Vulnerabilities and Countermeasures, PBX, TEM: Telecom Expense Management. C. Operating System Security Models: Operating System Models, Classic Security Models, Reference Monitor, Trustworthy Computing, International Standards for Operating System Security.	A) R, U B) U, AN, A C) U, EV
V	Comprehensive Security Framework: Virtualization, Application, and Physical Protection: A. Virtual Machines and Cloud Computing: Virtual Machines, Cloud Computing. B. Secure Application Design: Secure Development Lifecycle, Application Security Practices, Web Application Security, Client Application Security, Remote Administration Security. C. Physical Security: Classification of Assets, Physical Vulnerability Assessment, Choosing Site Location for Security, Securing Assets: Locks and Entry Controls, Physical Intrusion Detection.	A) U, AN B) A, AN C) AN, CR

List of Practical:		
Sr. No.	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	Configure Routers	A, CR
a	OSPF MD5 authentication.	
b	NTP	
c	to log messages to the syslog server.	
d	to support SSH connections.	
2.	Configure AAA Authentication	EC, CR
a	Configure a local user account on Router and configure authenticate on the console and vty lines using local AAA	
b	Verify local AAA authentication from the Router console and the PC client.	
3.	Configuring Extended ACLs	U, AN, EV

a	Configure, Apply and Verify an Extended Numbered ACL.	
b	Configure, Apply and Verify an Extended Named ACL.	
4.	Configure, Apply and Verify an Extended Named ACL.	U, AN, EV
a	Configure ACLs to mitigate attacks.	
b	Configure, Apply and Verify an IPv6 ACL.	
5.	Configuring a Zone-Based Policy Firewall (ZPF)	A, AN, EV
a	Configure a ZPF Firewall on Router.	
b	Verify ZPF firewall functionality using ping, SSH and web browser.	
6.	Configure IOS Intrusion Prevention System (IPS)	A, AN, EV
a	Enable IOS IPS.	
b	Modify an IPS signature.	
c	Verify IPS.	
7.	Layer 2 Security	U, A, EV
a	Assign the Central switch as the root bridge.	
b	Secure spanning-tree parameters to prevent STP manipulation attacks.	
c	Enable port security to prevent CAM table overflow attacks.	
8.	Layer 2 VLAN Security	U, A, EV
a	Enable trunking and configure security on trunk links.	
b	Create a new management VLAN.	
c	Implement an ACL to prevent outside users from accessing the management VLAN.	
9.	Configure and Verify a Site-to-Site IPsec VPN	
a	Configure a router to support a site-to-site IPsec VPN with another router.	A, AN, EV
10.	Configuring ASA Basic Settings and Firewall	A, AN, EV
a	Configure basic ASA settings and interface security levels.	
b	Configure routing, address translation and inspection policy.	
c	Configure DHCP, AAA, and SSH.	
d	Configure a DMZ, Static NAT, and ACLs.	

Learning Outcome:After completion of Course the students will be able to:

1. Contemplate that a comprehensive approach is the only way to be secure by applying right policies and philosophies.
2. Think beyond where information is stored and think more about how it is stored or more accurately how it is protected.
3. Comprehend how the underlying design of the network play and integral role in defining electronic boundaries and in enabling an organization to effectively protect, manage and secure access to information assets within that perimeter and outside of this perimeter.
4. Overcome the challenges faced in successful deploying an IDS or IPS and the helpfulness of operating system security model to an era of virtual machines and cloud security solutions.
5. Apply the important security considerations that should be part of the development cycle of web applications, client applications and remote administration illustrating security issues and how to solve them.

Books and References:

1. Rhodes-Ousley, M. (2013). The complete reference: Information security (2nd ed.). McGraw-Hill.
2. Dykstra, J. (2017). Essential cybersecurity science (5th ed.). O'Reilly.
3. Conklin, W. A., & White, G. (2019). Principles of computer security: CompTIA Security+ and beyond (2nd ed.). McGraw-Hill.
4. Stallings, W. (2017). Computer security: Principles and practice (4th ed.). Pearson.
5. Kaufman, C., Perlman, R., & Speciner, M. (2002). Network security: Private communication in a public world (2nd ed.). Prentice Hall.
6. Bezroukov, N. (2017). Cybersecurity and cyberwar: What everyone needs to know (1st ed.). Oxford University Press.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	10	10	20	30	20	10	100%

MCCSCT406 Business Intelligence

B. Sc (Information Technology)		Semester – VI		
Course Name: Business Intelligence		Course Code: MCCSCT406		
Vertical:		Major		
Periods per week (1 Period is 60 minutes)		3		
Practical per week (1 Period is 60 minutes)		2		
Credits		4		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Gain an understanding of the concept of Business Intelligence (BI), including its definition, working principles, and the realities associated with it.
2. Explore the reasons why businesses need BI, including its main uses, importance, and the benefits it offers. Understand the evolution of BI over time.
3. Familiarize yourself with the roles and responsibilities within a BI team and understand the typical structure of a BI team in an organization.
4. Identify ideas for success with BI, including strategies for management, end-users, and BI teams. Explore unconventional approaches to enhance the effectiveness of BI initiatives.
5. Introduce the concept of Integrated Business Intelligence (IBI) and provide guidance on how to initiate and embark on an IBI journey.

Module	Name	Lectures
I	What is Business Intelligence	9
II	The Essential Role of Business Intelligence	9
III	Evaluating Business Intelligence	9
IV	Business Intelligence for Success	9
V	Introduction to IBI	9
	Total	45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	A. What is Business Intelligence : Introduction, Working, Realities, Demystify coinage	A) U,R,AN
II	<u>The Essential Role of Business Intelligence</u> A. Why do Businesses Need BI?: Example, Main uses of BI, Importance of BI, Necessity of BI, ,Benefits of BI, Evolution of BI B. Types of Business Intelligence: BI types based on various parameters, BI types based on type of analytics, Varieties of BI implementation. C. Challenges in Business Intelligence: Main phases in a BI journey, Challenges faced in phases of BI.	A) U, R, AN B) U,R,AN C) U,R,AN
III	<u>Evaluating Business Intelligence</u> A. Roles in Business Intelligence: Typical BI team structure, BI roles and responsibilities B. Financials of Business Intelligence: Cost of BI, ROI for BI.	A) U, R, AN, EV B) U, R, AN, EV
IV	<u>Business Intelligence for Success</u> A. Ideas for Success with BI: Ideas for management, Ideas for BI users, B. Unconventional ideas for BI teams.	A) U,R,AN B) U,R,AN
V	<u>Introduction to IBI</u> A. Introduction to IBI: What is IBI? How to start with IBI? B. BI Architectures: BI architecture, Examples C. Demystify Tech, Tools and Concepts in BI: Technologies and tools, Concepts.	A)U, R, AN B) U, R, AN C) U, R, AN

List of Practicals		
Sr. No.	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1	Import the legacy data from different sources such as (Excel , SqlServer, Oracle etc.) and load in the target system. (You can download sample database such as Adventureworks, Northwind, foodmart etc.)	CR,AN,U,A
2	Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sqlserver.	CR,AN,U,A

3	a. Create the Data staging area for the selected database. b. Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.	CR,AN,U,A
4	a.Create the ETL map and setup the schedule for execution. b. Execute the MDX queries to extract the data from the datawarehouse.	CR,AN,U,A
5	a. Import the datawarehouse data in Microsoft Excel and create the Pivot table and Pivot Chart. b. Import the cube in Microsoft Excel and create the Pivot table and Pivot Chart to perform data analysis.	CR,AN,U,A
6	Apply the what – if Analysis for data visualization. Design and generate necessary reports based on the data warehouse data.	CR,AN,U,A
7	Perform the data classification using classification algorithm.	CR,AN,U,A
8	Perform the data clustering using clustering algorithm.	CR,AN,U,A
9	Perform the Linear regression on the given data warehouse data.	CR,AN,U,A
10	Perform the logistic regression on the given data warehouse data.	CR,AN,U,A

Learning Outcomes: After completion of the course students are supposed to be able to:

1. Develop a clear understanding of the term "Business Intelligence" and its implications in the business world.
2. Recognize the significance of BI in modern business environments and comprehend its potential to drive informed decision-making and improve overall organizational performance.
3. Gain insights into the various roles involved in implementing and managing BI initiatives, and appreciate the collaborative efforts required for successful BI implementation.
4. Develop innovative ideas and approaches to leverage BI effectively within an organization, considering the perspectives of management, users, and BI teams.
5. Understand the concept of IBI and gain insights into the architectural considerations and examples associated with integrating BI within a business context. Familiarize yourself with the technologies, tools, and core concepts relevant to BI implementation.

Books and References:

1. Kumar, A. V. K. (2022). Business intelligence demystified (1st ed.). BPB Publications.
2. Howson, C. (2007). Successful business intelligence: Secrets to making BI a killer app (1st ed.). McGraw-Hill.
3. Kimball, R., & Ross, M. (2013). The data warehouse toolkit: The definitive guide to dimensional modeling (3rd ed.). Wiley.
4. Davenport, T. H., & Harris, J. G. (2007). Competing on analytics: The new science of winning (1st ed.). Harvard Business Review Press.
5. Loshin, D. (2003). Business intelligence: The savvy manager's guide (1st ed.). Morgan Kaufmann.
6. Devlin, B. (1997). Data warehouse – From architecture to implementation (1st ed.). Addison-Wesley.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	10	10	20	100%

MCCSCT408 Advanced Mobile Programming Practical

B. Sc (Information Technology)		Semester – VI		
Course Name: Advanced Mobile Programming Practical		Course Code: MCCSCT408		
Vertical:		Major		
Practical per week (1 Period is 60 minutes)		2		
Credits		2		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course objective:

1. Understand the fundamental concepts and components of Android development, including activities, services, content providers, and broadcast receivers.
2. Gain proficiency in utilizing Android resources such as colors, themes, strings, drawables, and dimensions to enhance the visual aspects of an application.
3. Develop skills in programming activities and fragments, including understanding their life cycles and implementing multiple activities and fragments in an application.
4. Learn to design and implement different layouts, such as coordinate, linear, relative, table, absolute, frame, list view, and grid view, to create user interfaces in Android applications.
5. Explore and implement UI elements like app bars, fragments, and other UI components to enhance the user experience and interaction in Android applications.

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Practical No	Details	Level of Knowledge Applicable as per Blooms Taxonomy
0	Introduction to Android, Introduction to Android Studio IDE, Application Fundamentals: Creating a Project, Android Components, Activities, Services, Content Providers, Broadcast Receivers, Interface overview, Creating Android Virtual device, USB debugging mode, Android Application Overview. Simple “Hello World” program.	U, R, A
1	Programming Resources	CR, A, U
a.	Android Resources: (Color, Theme, String, Drawable, Dimension, Image)	
2	Programming Activities and fragments	CR, AN, A
a	Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.	

3	Programs related to different Layouts	CR, AN, R
a	Coordinate	
b	Linear	
c	Relative	
d	Table	
e	Absolute	
f	Frame	
g	List View	
h	Grid View	
4	Programming UI elements	CR, U
a	AppBar, Fragments, UI Components	
5	Programming menus, dialog, dialog fragments	CR, U, AN
6	Programs on Intents, Events, Listeners and Adapters	CR, U, A
a	The Android Intent Class, Using Events and Event Listeners	
7	Programs on Services, notification and broadcast receivers	CR, U, AN
8	DataBase	CR, U, AN, A
	a. Database Programming with SQLite	
	b. Programming Network Communications and Services (JSON)	
9	Programming threads, handles and asynchronized programs	CR, U, AN, A
10	Multi-Media	CR, U, AN
	a. Programming Media API and Telephone API	
	b. Programming Security and permissions	

Learning Outcomes: After completion of the course students are supposed to be able to:

1. Develop a solid understanding of Android application development and the Android Studio IDE, enabling the creation of basic Android projects and the utilization of essential Android components.
2. Apply Android resources effectively, including colors, themes, strings, drawables, and dimensions, to enhance the visual appeal and customization of Android applications.
3. Demonstrate proficiency in programming activities and fragments, including managing their life cycles, implementing multiple activities, and effectively utilizing fragments in Android applications.
4. Design and implement diverse layouts, such as coordinate, linear, relative, table, absolute, frame, list view, and grid view, to create visually appealing and user-friendly interfaces in Android applications.

5. Implement various UI elements, such as app bars, fragments, and other UI components, to enhance user interaction and provide a seamless user experience in Android applications.

Books & References:

1. Phillips, B., Stewart, C., & Marsicano, K. (2021). Android programming: The Big Nerd Ranch guide (4th ed.). Big Nerd Ranch.
2. Meier, R., & Lake, I. (2018). Professional Android (4th ed.). Wrox.
3. Prusty, N. (2018). Android development for beginners: Learn Android app development [eBook]. Udemy.
4. Howard, J. (2020). Android Studio 4.0 development essentials. Packt Publishing.
5. Boyd, J. (2020). Android programming for beginners. Packt Publishing.
6. Saha, M. (2021). Master Android development with Kotlin. Packt Publishing.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	10	30	10	10	10	30	100%

MCCSCT313 Principles of GIS

B. Sc (Information Technology)		Semester – VI		
Course Name: Principles of GIS		Course Code: MCCSCT313		
Vertical:		Minor		
Periods per week (1 Period is 60 minutes)		02		
Credits		02		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Introduce the nature of Geographic Information Systems (GIS) and their applications. Understand spatial data, models, and representations of the real world.
2. Explore data management and processing systems, including GIS software, architecture, spatial data infrastructure, and stages of spatial data handling.
3. Learn about spatial referencing and positioning, including reference surfaces, coordinate systems, map projections, satellite-based positioning, and data entry and preparation techniques.
4. Develop skills in spatial data analysis, including retrieval, classification, measurement, overlay functions, neighborhood functions, analysis techniques, and understanding error propagation in spatial data processing.
5. Understand data visualization in GIS, including the visualization process, strategies, cartographic toolbox, mapping qualitative and quantitative data, terrain elevation mapping, and map dissemination.

Module	Name	Lectures
I	A Gentle Introduction to GIS	6
II	Data Management and Processing Systems	6
III	Spatial Referencing and Positioning	6
IV	Spatial Data Analysis	6
V	Data Visualization	6
		30

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>A Gentle Introduction to GIS:</u> A. The nature of GIS: Some fundamental observations, Defining GIS, GISystems, GIScience and GIApplications, Spatial data and Geoinformation. B. The real world and representations of it: Models and modelling, Maps, Databases, Spatial databases and spatial analysis C. Geographic Information and Spatial Database D. Models and Representations of the real world E. Geographic Phenomena: Defining geographic phenomena, types of geographic phenomena, Geographic fields, Geographic objects, Boundaries F. Computer Representations of Geographic Information: Regular tessellations, irregular tessellations, Vector representations, Topology and Spatial relationships, Scale and Resolution, Representation of Geographic fields, Representation of Geographic objects G. Organizing and Managing Spatial Data H. The Temporal Dimension	A) R, U B) R, U C) R, U D) R, U E) R, U, A G) R, U H) R, A
II	<u>Data Management and Processing Systems:</u> A. Hardware and Software Trends B. Geographic Information Systems: GIS Software, GIS Architecture and functionality, Spatial Data Infrastructure (SDI) C. Stages of Spatial Data handling: Spatial data handling and preparation, Spatial Data Storage and maintenance, Spatial Query and Analysis, Spatial Data Presentation. D. Database management Systems: Reasons for using a DBMS, Alternatives for data management, The relational data model, Querying the relational database. E. GIS and Spatial Databases: Linking GIS and DBMS, Spatial database functionality.	A) R, U B) R, U C) R, U, A D) R, U, EV E) R, U, EV
III	<u>Spatial Referencing and Positioning:</u> A. Spatial Referencing: Reference surfaces for mapping, Coordinate Systems, Map Projections, Coordinate Transformations	A) U, A, AN

	B. Satellite-based Positioning: Absolute positioning, Errors in absolute positioning, Relative positioning, Network positioning, code versus phase measurements, Positioning technology C. Data Entry and Preparation D. Spatial Data Input: Direct spatial data capture, Indirect spatial data capture, Obtaining spatial data elsewhere E. Data Quality: Accuracy and Positioning, Positional accuracy, Attribute accuracy, temporal accuracy, Lineage, Completeness, Logical consistency F. Data Preparation: Data checks and repairs, Combining data from multiple sources G. Point Data Transformation: Interpolating discrete data, Interpolating continuous data	B) U, A, R C) U, A, AN D) U, A, R E) U, R, EV F) U, R, EV G) U, R
IV	Spatial Data Analysis: A. Classification of analytical GIS Capabilities B. Retrieval, classification and measurement: Measurement, Spatial selection queries, Classification C. Overlay functions: Vector overlay operators, Raster overlay operators D. Neighbourhood functions: Proximity computations, Computation of diffusion, Flow computation, Raster based surface analysis E. Analysis: Network analysis, interpolation, terrain modeling F. GIS and Application models: GPS, Open GIS Standards, GIS Applications and Advances G. Error Propagation in spatial data processing: How Errors propagate, Quantifying error propagation	A) U, R, EV B) U, R, AN C) EV D) U, A E) A, AN F) A, U G) A, U, EV
V	Data Visualization: A. GIS and Maps, The Visualization Process B. Visualization Strategies: Present or explore? C. The cartographic toolbox: What kind of data do I have? How can I map my data? D. How to map? How to map qualitative data, How to map quantitative data, How to map the terrain elevation, How to map time series E. Map Cosmetics, Map Dissemination	A) U, A, AN B) U, A, R C) U, A, AN D) U, A, R E) U, R, EV

Practical No	Details	Level of Knowledge Applicable as per Blooms Taxonomy
0	Familiarizing Quantum GIS: Installation of QGIS, datasets for both Vector and Raster data, Maps.	CR, U
1	Creating and Managing Vector Data: Adding vector layers, setting properties, formatting, calculating line lengths and statistics	CR, U, A
2	Exploring and Managing Raster data: Adding raster layers, raster styling and analysis, raster mosaicking and clipping	CR, AN
3	Making a Map, Working with Attributes, Importing Spreadsheets or CSV files Using Plugins, Searching and Downloading OpenStreetMap Data	CR, AN, A
4	Working with attributes, terrain Data	CR, A, AN
5	Working with Projections and WMS Data	CR, A, U
6	Georeferencing Topo Sheets and Scanned Maps Georeferencing Aerial Imagery Digitizing Map Data	CR, A
7	Managing Data Tables and Spatial data Sets: Table joins, spatial joins, points in polygon analysis, performing spatial queries	CR, AN
8	Advanced GIS Operations 1: Nearest Neighbor Analysis, Sampling Raster Data using Points or Polygons, Interpolating Point Data	CR, U, AN
9	Advance GIS Operations 2: Batch Processing using Processing Framework Automating Complex Workflows using Processing Modeler Automating Map Creation with Print Composer Atlas	CR, AN
10	Validating Map data	CR, A, AN

Learning Outcomes: After completion of the course students are supposed to be able to:

1. Gain a comprehensive understanding of GIS fundamentals, including the nature of GIS, spatial data, models, representations, and the real-world applications of GIS.
2. Develop proficiency in data management and processing systems, including the use of GIS software, understanding GIS architecture, spatial data infrastructure, and performing spatial data handling tasks.
3. Acquire knowledge and skills in spatial referencing and positioning, including the use of reference surfaces, coordinate systems, map projections, satellite-based positioning, and data entry and preparation techniques ensuring data quality.
4. Demonstrate proficiency in spatial data analysis, including retrieval, classification, measurement, overlay functions, neighborhood functions, analysis techniques, and understanding the propagation of errors in spatial data processing.
5. Apply effective data visualization strategies in GIS, including understanding the visualization process, utilizing the cartographic toolbox, mapping qualitative and quantitative data, terrain elevation mapping, and effectively disseminating maps.

Books and References:

1. Huisman, O., & Rolf, A. (2009). Principles of geographic information systems: An introductory text book (4th ed.). The International Institute of Geoinformation Science and Earth Observation.
2. Burrough, P. A., & McDonnell, R. A. (1999). Principles of geographic information systems (3rd ed.). Oxford University Press.
3. Laurini, R., & Thompson, D. (1994). Fundamentals of spatial information systems. Academic Press.
4. Demers, M. N. (2009). Fundamentals of geographic information systems (4th ed.). Wiley Publications.
5. Chang, K.-t. (2013). Introduction to geographic information systems (7th ed.). McGraw-Hill.
6. Bolstad, P. (2016). GIS fundamentals: A first text on geographic information systems (5th ed.). XanEdu Publishing Inc.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	30	10	10	5	25	100%

MCCSCT314 Cloud Computing Fundamentals

B. Sc (Information Technology)		Semester – VI		
Course Name: Cloud Computing Fundamentals		Course Code: MCCSCT314		
Vertical:		Minor		
Periods per week (1 Period is 60 minutes)		02		
Credits		02		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Understand the concepts and technologies related to distributed systems and scalable computing over the internet.
2. Gain knowledge of design principles, job/resource management, and virtualization techniques in computer clusters and data centers.
3. Explore Microsoft Azure services, subscriptions, and resource management using Azure Resource Manager.
4. Learn about storage management in Azure, including different storage types and configuring storage accounts.
5. Develop skills in deploying and configuring Azure virtual machines, considering architectural considerations and cost-saving opportunities.

Module	Name	Lectures
I	Design and Optimization of Scalable Distributed Systems: From Cloud Computing to Computer Clusters for High-Performance Parallelism	6
II	Design and Management of Virtualized Computer Clusters for Scalable and Efficient Data Center Operations	6
III	Managing and Optimizing Cloud Resources with Microsoft Azure: An Introduction to Azure Services and Resource Management Tools	6
IV	Building and Managing Scalable Infrastructure on Microsoft Azure: Storage Solutions and Network Topology	6
V	Deploying and Managing Virtual Machines on Microsoft Azure: Best Practices for Configuration, Cost Optimization, and Scalability	6
	Total	30

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus	Lectures
I	<u>Design and Optimization of Scalable Distributed Systems: From Cloud Computing to Computer Clusters for High-Performance Parallelism:</u> A. IKS in Cloud Computing: Nalanda and Takshashila, Dharma, Vedic Texts / Mantras, Mahabharata's decisionmaking models, Ancient Indian manuscripts. B. Distributed System Models and Enabling Technologies: Scalable Computing Over the Internet, Technologies for Network-Based Systems, System Models for Distributed and Cloud Computing, Software Environments for Distributed Systems and Clouds, Performance, Security, and Energy Efficiency, Bibliographic Notes and Homework Problems. C. Computer Clusters for Scalable Parallel Computing: Clustering for Massive Parallelism, Computer Clusters and MPP Architectures,	A) C,R,AN B) U, R C) U, R
II	<u>Design and Management of Virtualized Computer Clusters for Scalable and Efficient Data Center Operations:</u> A. Design Principles, Job and Resource management of computer Cluster: Design Principles of Computer Clusters, Cluster Job and Resource Management, Case Studies of Top Supercomputer Systems. B. Virtual Machines and Virtualization of Clusters and Data Centers: Implementation Levels of Virtualization, Virtualization Structures/Tools and Mechanisms, Virtualization of CPU, Memory, and I/O Devices, Virtual Clusters and Resource Management, Virtualization for Data-Center Automation	A) U, R B) U, R
III	<u>Managing and Optimizing Cloud Resources with Microsoft Azure: An Introduction to Azure Services and Resource Management Tools:</u> A. Introducing Microsoft Azure: Microsoft Azure Services, Subscriptions, Staying current with Azure. B. Exploring Azure Resource Manager: Introduction to Azure resource manager, Azure regions, Azure Management tools, Azure CLI on desktop	A) U, R B) U, R
IV	<u>Building and Managing Scalable Infrastructure on Microsoft Azure: Storage Solutions and Network Topology:</u> A. Managing Storage in Azure: Azure storage data types, working with storage account, Azure disk storage B. Planning Network Topology: Understanding network components, creating virtual networks, configuring virtual networks, connecting virtual networks, deploying azure firewall, virtual private networks and gateways	A) U, R B) U, R

V	<u>Deploying and Managing Virtual Machines on Microsoft Azure: Best Practices for Configuration, Cost Optimization, and Scalability:</u> A. Deploying and Configuring Azure Virtual Machines: Planning VM deployment, recognising Azure VM components, Architectural Considerations, cost saving opportunities, deploying Azure VMs from Azure Marketplace, Configuring VMs, Starting, stopping and resizing VMS, Using Azure virtual desktop.	A) U, R
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Learning Outcomes: After completion of the course students are supposed to be able to:

1. Apply knowledge of distributed system models and technologies to design scalable computing solutions for internet-based systems.
2. Demonstrate proficiency in designing and managing computer clusters, including job/resource management and virtualization techniques.
3. Utilize Microsoft Azure services effectively, including Azure Resource Manager, for resource provisioning and management.
4. Manage storage in Azure by understanding different storage types and effectively working with storage accounts.
5. Successfully deploy and configure Azure virtual machines, considering architectural considerations, cost-saving opportunities, and utilizing Azure virtual desktop.

Books and References:

1. Hwang, K., Fox, G. C., & Dongarra, J. J. (2019). Distributed and cloud computing: From parallel processing to the Internet of Things. Elsevier.
2. Erl, T. (2013). Cloud computing: Concepts, technology & architecture (1st ed.). Prentice Hall.
3. Buyya, R., Vecchiola, C., & Selvi, S. T. (2013). Mastering cloud computing: Foundations and applications programming (1st ed.). McGraw-Hill.
4. Erl, T., Mahmood, Z., & Arenas, R. P. (2013). Cloud computing design patterns (1st ed.). Prentice Hall.
5. Smith, J. E., & Nair, R. (2005). Virtual machines: Versatile platforms for systems and processes (1st ed.). Morgan Kaufmann.
6. Lazić, B. (2012). Cloud computing for science and engineering (1st ed.). Wiley.
7. Mahadevan, B., Vaidhyasubramaniam, S., & Balasubramanian, K. (2022). Introduction to Indian Knowledge System: Concepts and Applications. PHI Learning.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	30	10	10	5	25	100%

MCCSCT315 IT Service Management

B. Sc (Information Technology)		Semester – VI		
Course Name: IT Service Management		Course Code: MCCSCT315		
Vertical:		Minor		
Periods per week (1 Period is 60 minutes)		02		
Credits		02		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Understand the fundamentals of IT service management and the ITIL 4 framework.
2. Explore the various management practices in ITIL, including general management, service management, and technical management.
3. Develop an understanding of driving stakeholder value and creating value for stakeholders.
4. Gain knowledge of creating, delivering, and supporting services within the ITIL service value system.
5. Explore the concept of High-Velocity IT and the importance of continual improvement.

Module	Name	Lectures
I	The ITIL Service Value System and Its Four Dimensions of Service Management	6
II	ITIL management practices	6
III	Drive Stakeholder Value	6
IV	Value Streams in Service Management: Creating, Delivering, and Supporting New Services	6
V	High-Velocity IT and continual improvement	6
	Total	30

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Details	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>The ITIL Service Value System and Its Four Dimensions of Service Management:</u>	

	A. Introduction: IT service management in the modern world, About ITIL 4, the structure and benefits of the ITIL 4 framework. B. Key concepts of service management: Value and value co-creation, Organizations, service providers, service consumers, and other stakeholders, Products and services, Service relationships, Value: outcomes, costs, and risks. C. The four dimensions of service management: Organizations and people, Information and technology, Partners and suppliers, Value streams and processes, External factors. D. The ITIL service value system: Service value system overview, Opportunity, demand, and value, The ITIL guiding principles, Governance, Service value chain, Continual improvement.	A) U, R, AN B) U, R, AN C) U, R, AN D) U, R, AN
II	<u>ITIL management practices:</u> A. ITIL management practices: General management practices, Service management practices, technical management practices.	A)U, AN, EV
III	<u>Drive Stakeholder Value:</u> A. Drive Stakeholder Value: Introduction, the customer journey, Step 1: Explore, Step 2: Engage, Step 3: Offer, Step 4: Agree, Step 5: Onboard, Step 6: Co-create, Step 7: Realize.	A) U, R, AN
IV	<u>Value Streams in Service Management: Creating, Delivering, and Supporting New Services:</u> A. Create, Delivery and Support (CDS): Service value system key concepts and challenges, using a shift-left approach, Plan and manage resources in the service value system, The use and value of technology across the service value system. B. Value streams for new services: Reviewing service value chains and service value streams, ITIL practices and value streams for new services, Change enablement, Service design and software development and management, Service validation and testing, Release management and deployment management C. Value streams for user support: ITIL practices and value streams for user support, Service desk, Incident management, Problem management, Knowledge management, Service level management, Monitoring and event management. D. How to create, deliver and support services	A) U, R, AN B) U, R, EV C)U, R, AN D) U, R, AN

V	<u>High-Velocity IT and Continual improvement:</u> A. High-Velocity IT: Introduction, Key concepts, culture, techniques, Continual improvement.	A) U, R, AN
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Learning Outcomes: After completion of the course students are supposed to be able to:

1. Apply the principles and concepts of ITIL 4 to enhance IT service management practices.
2. Demonstrate proficiency in implementing ITIL management practices.
3. Successfully drive stakeholder value and engage in value co-creation.
4. Develop the skills to create, deliver, and support services within the ITIL service value system.
5. Understand the principles and techniques of High-Velocity IT and foster a culture of continual improvement.

Books and References:

1. Axelos. (2011). ITIL lifecycle suite (5th ed.). TSO, AXELOS.
2. Van Bon, J. (2012). Foundations of IT service management with ITIL 2011 (2nd ed.). Van Haren Publishing.
3. Hollis, M. (2019). ITIL 4 essentials: Your essential guide for the ITIL 4 Foundation exam and beyond (1st ed.). TSO, AXELOS.
4. Grigoriu, A. (2011). ITIL V3 foundation handbook (3rd ed.). TSO, AXELOS.
5. Whitworth, G. (2020). ITIL 4 foundation exam study guide (1st ed.). Wiley.
6. Cannon, D. (2014). ITIL foundation exam study guide (3rd ed.). Wiley.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	10	10	20	100%

MCCSCT316 Cyber Laws and Patents

B. Sc (Information Technology)		Semester – VI		
Course Name: Cyber Laws and Patents		Course Code: MCCSCT316		
Vertical:		Minor		
Periods per week (1 Period is 60 minutes)		02		
Credits		02		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Critically analyze the power of arrest without a warrant under the IT Act, 2000 and its implications.
2. Understand the legal aspects of contracts in the Infotech world and jurisdiction in the cyber world.
3. Explore the battle against cyber squatters, copyright protection, and the legal issues surrounding content on the internet.
4. Examine the challenges and complexities of e-commerce taxation in the virtual world and the role of digital signatures, certifying authorities, and e-governance.
5. Gain knowledge of the legal aspects related to electronic records as evidence, the protection of cyber consumers, and the amendments in the Indian IT Act, 2000.

Module	Name	Lectures
I	Arrest Powers and Cyber Crime Legal Procedures under the IT Act, 2000	6
II	Legal Issues in Infotech Contracts and Cyber Jurisdiction	6
III	Combating Cyber Squatting and Ensuring Copyright Protection in the Digital World.	6
IV	Taxation in E-Commerce and Digital Signatures in E-Governance	6
V	Electronic Evidence and Cyber Consumer Protection in India	6
	Total	30

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR - Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>Arrest Powers and Cyber Crime Legal Procedures under the IT Act, 2000:</u> A. Power of Arrest Without Warrant Under the IT Act, 2000: A Critique, Crimes of this Millennium, Section 80 of the IT Act, 2000 – A Weapon or a Farce? Forgetting the Line Between Cognizable and Non-Cognizable Offences, Necessity of Arrest without Warrant from Any Place, Public or Otherwise, Check and Balances Against Arbitrary Arrests, Arrest for “About to Commit” an Offence Under the IT Act: A Tribute to Draco, Arrest, But NO Punishment! B. Cyber Crime and Criminal Justice: Penalties, Adjudication and Appeals Under the IT Act, 2000: Concept of “Cyber Crime “ and the IT Act , 2000, Hacking, Teenage Web Vandals, Cyber Fraud and Cyber Cheating, Virus on the Internet, Defamation, Harassment and E-mail Abuse, Cyber Pornography, Other IT Act Offences, Monetary Penalties, Adjudication and Appeals Under IT Act , 2000, Network Service Providers, Jurisdiction and Cyber Crime, Nature of Cyber Criminality, Strategies to Tackle Cyber Crime and Trends, Criminal Justice in India and Implications on Cyber Crime.	A) U, R, AN B) U, R, AN
II	<u>Legal Issues in Infotech Contracts and Cyber Jurisdiction:</u> A. Contracts in the Infotech World: Contracts in the Infotech World, Click-Wrap and Shrink-Wrap Contract: Status under the Indian Contract Act, 1872, Contract Formation Under the Indian Contract Act, 1872, Contract Formation on the Internet, Terms and Conditions of Contracts. B. Jurisdiction in the Cyber World: Questioning the Jurisdiction and Validity of the Present Law of Jurisdiction, Civil Law of Jurisdiction in India, Cause of Action, Jurisdiction and the Information Technology Act,2000, Foreign Judgements in India, Place of Cause of Action in Contractual and IPR Disputes, Exclusion Clauses in Contracts, Abuse of Exclusion Clauses, Objection of Lack of Jurisdiction, Misuse of the Law of Jurisdiction, Legal Principles on Jurisdiction in the United State of America, Jurisdiction Disputes w.r.t. the Internet in the United State of America.	A) U, AN, EV B) U, R, AN

III	<u>Combating Cyber Squatting and Ensuring Copyright Protection in the Digital World:</u> A. Battling Cyber Squatters and Copyright Protection in the Cyber World: Concept of Domain Name and Reply to Cyber Squatters, Meta-Tagging, Legislative and Other Innovative Moves Against Cyber Squatting, The Battle Between Freedom and Control on the Internet, Works in Which Copyright Subsists and meaning of Copyright, Copyright Ownership and Assignment, License of Copyright, Copyright Terms and Respect for Foreign Works, Copyright Infringement, Remedies and Offences, Copyright Protection of Content on the Internet; Copyright Notice, Disclaimer and Acknowledgement, Downloading for Viewing Content on the Internet, Hyper-Linking and Framing, Liability of ISPs for Copyright Violation in the Cyber World: Legal Developments in the US, Napster and its Cousins: A Revolution on the Internet but a Crisis for Copyright Owners, Computer Software Piracy.	A) U, R, AN
IV	<u>Taxation in E-Commerce and Digital Signatures in E-Governance:</u> A. E-Commerce Taxation: Real Problems in the Virtual World: A Tug of War on the Concept of ‘Permanent Establishment’, Finding the PE in Cross Border E-Commerce, The United Nations Model Tax Treaty, The Law of Double Taxation Avoidance Agreements and Taxable Jurisdiction Over Non-Residents, Under the Income Tax Act, 1961, Tax Agents of Non-Residents under the Income Tax Act, 1961 and the Relevance to E-Commerce, Source versus Residence and Classification between Business Income and Royalty, The Impact of the Internet on Customer Duties, Taxation Policies in India: At a Glance. B. Digital Signature, Certifying Authorities and E-Governance: Digital Signatures, Digital Signature Certificate, Certifying Authorities and Liability in the Event of Digital Signature Compromise, E-Governance in India: A Warning to Babudom!	A) U, R, AN B) U, R, EV
V	<u>Electronic Evidence and Cyber Consumer Protection in India:</u> A. The Indian Evidence Act of 1872 v. Information Technology Act, 2000: Status of Electronic Records as Evidence, Proof and Management of Electronic Records; Relevancy, Admissibility and Probative Value of E-Evidence, Proving Digital Signatures, Proof of Electronic Agreements, Proving Electronic Messages, Other Amendments in the Indian Evidence Act by the IT	A) U, R, AN B) U, R, AN

	Act, Amendments to the Bankers Books Evidence Act, 1891 and Reserve Bank of India Act, 1934. B. Protection of Cyber Consumers in India: Are Cyber Consumers Covered Under the Consumer Protection Act? Goods and Services, Consumer Complaint, Defect in Goods and Deficiency in Services, Restrictive and Unfair Trade Practices, Instances of Unfair Trade Practices, Reliefs Under CPA, Beware Consumers, Consumer Foras, Jurisdiction and Implications on cyber-Consumers in India, Applicability of CPA to Manufacturers, Distributors, Retailers and Service Providers Based in Foreign Lands Whose Goods are Sold or Services Provided to a Consumer in India. Amendments in Indian IT Act 2000.	
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Learning Outcomes: After completion of the course students are supposed to be able to:

1. Evaluate the provisions and limitations of the power of arrest without a warrant under the IT Act, 2000 and its impact on individuals' rights.
2. Apply the legal principles governing contracts in the Infotech world and understand the jurisdictional issues in the cyber world.
3. Analyze the strategies and legal measures against cyber squatters, copyright infringement, and protection of content on the internet.
4. Assess the challenges and implications of e-commerce taxation in the virtual world and understand the role of digital signatures, certifying authorities, and their impact on e-governance.
5. Demonstrate understanding of the legal requirements for electronic records as evidence, the protection of cyber consumers, and the amendments in the Indian IT Act, 2000.

Books and References:

1. Sood, V. (2001). Cyber law simplified. TMH Education.
2. Kosseff, J. (2017). Cybersecurity law. Wiley.
3. Brenner, S. W. (2010). Cybercrime: Criminal threats from cyberspace. Praeger.
4. Solove, D. J., & Schwartz, P. M. (2020). Information privacy law (6th ed.). Aspen Publishers.
5. Lipton, J. D. (2014). Rethinking cyberlaw: A new vision for the Internet age. Edward Elgar Publishing.
6. Trappe, W., & Washington, L. C. (2006). Introduction to cryptography with coding theory (2nd ed.). Pearson Education. (useful in understanding legal implications of encryption and data protection)
7. Spinello, R. A. (2014). Cyberethics: Morality and law in cyberspace (5th ed.). Jones & Bartlett Learning.
8. Jasanoff, S. (Ed.). (2007). Reframing rights: Bioconstitutionalism in the genetic age. MIT Press. (covers legal theory related to patents and genetic information)
9. Holyoak, J., & Torremans, P. (2013). Intellectual property law (6th ed.). Oxford University Press.
10. WIPO. (2020). Understanding patents. World Intellectual Property Organization.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	10	10	20	100%

MCCSCT407 EARN

B. Sc (Information Technology)		Semester – VI		
Course Name: EARN		Course Code: MCCSCT407		
Vertical:		VSEC		
Periods per week (1 Period is 60 minutes)		03		
Practical per week (1 Period is 60 minutes)		02		
Credits		04		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objective:

1. Understand the Node environment, including installation, hosting environments, and Node C/C++ Add-ons.
2. Explore Node's building blocks, such as global objects, events, asynchronous event handling, and the basics of Node modules and npm.
3. Introduce React and its foundational concepts, including React elements, components, JSX, and functional programming with JavaScript.
4. Provide an introduction to TypeScript and cover core concepts of Angular, including modules, components, directives, and decorators.
5. Cover advanced topics in Node and React, including state management with hooks, routing, forms, HTTP, and dependency injection in Angular.

Unit/Module	Name	Lectures
I	Mastering Node.js: From Installation and Basics to Web Development and Advanced Concepts	9
II	Comprehensive Node.js Development: From Local Systems and Networking to Full-Stack Applications and ES6 Features	9
III	Mastering React: Foundations, State Management, Routing, and Data Handling	9
IV	Mastering Angular: TypeScript Fundamentals, Core Concepts, and Essential Features	9
V	Advanced Angular: Forms, HTTP Handling, and Routing Techniques	9
		45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>Mastering Node.js: From Installation and Basics to Web Development and Advanced Concepts</u> A) The Node Environment: Installing Node, Saying Hello to the World with Node, Node Hosting Environments, The Node LTS and Upgrading Node, Node, V8, and ES6 Advanced: Node C/C++ Add-ons B) Node Building Blocks: Global Objects, Events, and Node's Asynchronous Nature: The global and process Objects, Buffers, Typed Arrays, and Strings, Node's Callback and Asynchronous Event Handling, Nested Callbacks and Exception Handling C) Basics of Node Modules and Node Package Manager (npm): An Overview of the Node Module System, An In-Depth Exploration of NPM, Creating and Publishing Your Own Node Module, Discovering Node Modules and Three Must-Have Modules D) Node and the Web: The HTTP Module: Server and Client, What's Involved in Creating a Static Web Server, Using Apache to Proxy a Node Application, Parsing the Query with Query String, DNS Resolution	A) U,R,A B)U, R, A, AN, CR C)U,R,A,AN C)U, R, A, AN, CR
II	<u>Comprehensive Node.js Development: From Local Systems and Networking to Full-Stack Applications and ES6 Features</u> A) Node and the Local System: Exploring the Operating System, Streams and Pipes, A Formal Introduction to the File System, Resource Access with Path, Creating a Command-Line Utility, Compression/Decompression with ZLib, Pipes and ReadLine B) Networking, Sockets, and Security: Servers, Streams, and Sockets, Guards at the Gate C) Child Processes: child_process.spawn, Running a Child Process Application in Windows D) Node and ES6: Strict Mode, let and const, Arrow Functions, Classes, Promises with Bluebird E) Full-Stack Node Development: The Express Application Framework, MongoDB and Redis Database Systems, AngularJS and Other Full-Stack Frameworks	A) U, R, AN, A B) U, R, AN, A C)R, U, A, AN D)U, R, AN, A
III	<u>Mastering React: Foundations, State Management, Routing, and Data Handling</u> B) Introduction to React: A Strong Foundation, React's Past and Future, React Developer Tools, JavaScript for React, Functional Programming with JavaScript	A) R, U, A, AN B) R, U, A, CR

	C) How React Works: Page Setup, React Elements, ReactDOM, React Components, React with JSX, Mapping Arrays with JSX, React Fragments. D) React State Management & Hooks: The useState Hook, State in Component Trees, Sending State Down a Component Tree, Sending Interactions Back up a Component, Using Refs, useContext, useEffect, useReducer E) Routing, Incorporating Data & Error Handling : React Router, Requesting Data, Render Props, Suspense, Error Boundaries	C) U, R, AN, A D) R, U, AN, A
IV	<u>Mastering Angular: TypeScript Fundamentals, Core Concepts, and Essential Features</u> A) Introduction: Intro to TypeScript, Template, Angular Modules, Bootstrapping, Component Tree, Classes, Looping, Property & Event Binding B) Core : Angular CLI, Lifecycle Hooks, ViewChildren & ContentChildren, Components, Built-in Directives, Decorators, Modules, Pipes	A) R, U, AN, A, CR B) R, U, AN, A
V	<u>Advanced Angular: Forms, HTTP Handling, and Routing Techniques</u> A) Forms: Template Driven Forms, Reactive Model Form, Dependency Injection & Providers: Injectors, Provider, B) HTTP: Core HTTP API, Promises, Handling errors, Headers, Observables C) Routing: Route Configuration, Navigation, Parameterised Routes, Nested Routes, Router Guards	A) R, U, A, AN B) R, U, AN, A, CR C) R, U, AN, A, CR

List of Practical:		
Sr. No	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
1.	Create Chat application by using Socket.IO	U, A, AN, CR
2.	a. Perform CRUD operations by using Express with MongoDB b. Perform File(image/doc) Upload operation by using MongoDB	U, A, AN, CR

3.	Creating and Publishing Your Own Node Module	U, A, AN
4.	b) Create TODO list by using react hooks with different components. c) Create and validate the user form in React.	U, A, AN, CR
5.	Fetch the API data and render the list on React app.	U, A, AN, CR
6.	In Angular create TODO list with different components.	U, A, AN, CR
7.	In Angular create TODO list with different components.	U, R, A, AN, CR
8.	Create and validate the user form in Angular (Reactive Form/Template driven Form).	U, R, A, AN, CR
9.	Fetch the API data and render the list on Angular app.	U, A, AN, CR
10.	In Angular create and update the user data by using API.	U, A, AN, CR

Learning Outcomes: After completion of Course, the learners will be able to:

1. Gain proficiency in working with the Node environment, including installing Node, using global objects, handling events, and utilizing Node modules and npm.
2. Develop skills in working with streams, pipes, the file system, networking, and security in Node. Understand how to create command-line utilities and compress/decompress data.
3. Acquire knowledge and practical experience in React, including understanding React elements, components, JSX, and state management with hooks.
4. Master the fundamentals of Angular, including TypeScript, Angular CLI, components, modules, directives, and lifecycle hooks.
5. Demonstrate proficiency in working with forms, HTTP, routing, and dependency injection in Angular. Understand how to handle data, errors, and implement route guards.

Books and References:

1. Powers, S. (2016). Learning Node: Moving to the server side (2nd ed.). O'Reilly.
2. Hughes-Croucher, T., & Wilson, M. (2009). Node: Up and running (1st ed.). O'Reilly.
3. Copes, F. (n.d.). The NodeJS handbook.
4. Banks, A., & Porcello, E. (2020). Learning React: Modern patterns for developing React apps (2nd ed.). O'Reilly.
5. Hussain, A. (2016). Angular from theory to practice (3rd ed.). Code Craft.
6. Freeman, A. (2020). Pro Angular (5th ed.). Apress.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	15	10	15	100%

MCCSCT312 Linux Administration

B. Sc (Information Technology)		Semester – VI		
Course Name: Linux Administration		Course Code: MCCSCT312		
Vertical:		VSEC		
Periods per week (1 Period is 60 minutes)		03		
Practical per week (1 Period is 60 minutes)		02		
Credits		04		
Evaluation System		Duration (in Hours)	Total Marks	Minimum Passing Marks
Theory	Continuous Internal Assessment	--	40	16
	End Semester Examination	2	60	24
Practical	Continuous Internal Assessment	--	20	8
	End Semester Examination	2	30	12

Course Objectives:

1. Introduce Red Hat Enterprise Linux and the duties of a Linux system administrator. Familiarize with the Bash shell and basic file system management tasks.
2. Configure and manage storage, including partitions, logical volumes, file systems, snapshots, and encrypted volumes. Learn about networking, services, and runlevels.
3. Understand user and group management, including password management, external authentication sources, permissions, and access control lists.
4. Secure the server using iptables, including firewall configuration, advanced iptables settings, and cryptographic services for SSL and GPG.
5. Configure server features such as file sharing with NFS and Samba, DNS and DHCP services, setting up a mail server, configuring Apache web server, and Bash shell scripting.

Module	Name	Lectures
I	Mastering Red Hat Enterprise Linux: System Administration, Command Line, and Software Management	9
II	Configuring and Managing Linux Storage, Networking, and User Permissions	9
III	Securing, Cryptographic Services, and File Sharing Configuration on Linux Servers	9
IV	Configuring DNS, DHCP, Mail, and Apache Servers on Red Hat Enterprise Linux	9
V	Bash Scripting, High-Availability Clustering, and Automated Installation on Red Hat Linux	9
		45

R- Remember, U- Understand, A – Apply, AN- Analyze, EV- Evaluate, CR – Create

Unit	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
I	<u>Mastering Red Hat Enterprise Linux: System Administration, Command Line, and Software Management</u> A) Introduction to Red Hat Enterprise Linux: Linux, Open Source and Red Hat, Origins of Linux, Distributions, Duties of Linux System Administrator. B) Command Line: Working with the Bash Shell, Getting the Best of Bash, Useful Bash Key Sequences, Working with Bash History, Performing Basic File System Management Tasks, Working with Directories, Piping and Redirection, Finding Files C) System Administration Tasks: Performing Job Management Tasks, System and Process Monitoring and Management, Managing Processes with ps, Sending Signals to Processes with the kill Command, using top to Show Current System Activity, Managing Process Niceness, Scheduling Jobs, Mounting Devices, Working with Links, Creating Backups, Managing Printers, Setting Up System Logging, Setting Up Rsyslog, Common Log Files, Setting Up Logrotate D) Managing Software: Understanding RPM, Understanding Meta Package Handlers, Creating Your Own Repositories, Managing Repositories, Installing Software with Yum, Querying Software, Extracting Files from RPM Packages	A)U, R, A, AN, CR B)U, R, AN C)U,R,AN, A D)U, R, AN, A, CR
II	<u>Configuring and Managing Linux Storage, Networking, and User Permissions</u> A) Configuring and Managing Storage: Understanding Partitions and Logical Volumes, Creating Partitions, Creating File Systems, File Systems Overview, Creating File Systems, Changing File System Properties, Checking the File System Integrity, Mounting File Systems Automatically Through fstab, Working with Logical Volumes, Creating Logical Volumes, Resizing Logical Volumes, Working with Snapshots, Replacing Failing Storage Devices, Creating Swap Space, Working with Encrypted Volumes B) Connecting to the Network: Understanding NetworkManager, Working with Services and Runlevels, Configuring the Network with NetworkManager, Working with system-config-network, NetworkManager Configuration Files, Network Service Scripts, Networking from the Command Line, Troubleshooting Networking, Setting Up IPv6, Configuring SSH, Enabling the SSH	A) U, R, AN, A B) U, R, AN, A

	Server, Using the SSH Client, Using PuTTY on Windows Machines, Configuring Key-Based SSH Authentication, Using Graphical Applications with SSH, Using SSH Port Forwarding, Configuring VNC Server Access C) Working with Users, Groups, and Permissions: Managing Users and Groups, Commands for User Management, Managing Passwords, Modifying and Deleting User Accounts, Configuration Files, Creating Groups, Using Graphical Tools for User, and Group Management, Using External Authentication Sources, the Authentication Process, sssd, nsswitch, Pluggable Authentication Modules, Managing Permissions, the Role of Ownership, Basic Permissions: Read, Write, and Execute, Advanced Permissions, Working with Access Control Lists, Setting Default Permissions with umask, Working with Attributes	C) U, R, AN,
III	<u>Securing, Cryptographic Services, and File Sharing Configuration on Linux Servers</u> A) Securing Server with iptables: Understanding Firewalls, Setting Up a Firewall with system-config-firewall, Allowing Services, Trusted Interfaces, Masquerading, Configuration Files, Setting Up a Firewall with iptables, Tables, Chains, and Rules, Composition of Rule, Configuration Example, Advanced iptables Configuration, Configuring Logging, The Limit Module, Configuring NAT B) Setting Up Cryptographic Services: Introducing SSL, Proof of Authenticity: The Certificate Authority, Managing Certificates with openssl, Creating a Signing Request, Working with GNU Privacy Guard, Creating GPG Keys, Key Transfer, Managing GPG Keys, Encrypting Files with GPG, GPG Signing, Signing RPM Files C) Configuring Server for File Sharing: What is NFS? Advantages and Disadvantages of NFS, Configuring NFS4, Setting Up NFSv4, Mounting an NFS Share, Making NFS Mounts Persistent, Configuring Automount, Configuring Samba, Setting Up a Samba File Server, Samba Advanced Authentication Options, Accessing Samba Shares, Offering FTP Services.	A) R, U, A, AN B) R, U, A, CR C) U, R, AN, A
IV	<u>Configuring DNS, DHCP, Mail, and Apache Servers on Red Hat Enterprise Linux</u> A) Configuring DNS and DHCP: Introduction to DNS, The DNS Hierarchy, DNS Server Types, The DNS Lookup Process, DNS Zone Types, Setting Up a DNS Server, Setting Up a Cache-Only Name Server, Setting Up a Primary Name Server, Setting Up a Secondary Name Server, Understanding DHCP, Setting Up a DHCP Server B) Setting Up a Mail Server: Using the Message Transfer Agent, the Mail Delivery Agent, the Mail User Agent,	A) R, U, AN, A, CR

	Setting Up Postfix as an SMTP Server, Working with Mutt, Basic Configuration, Internet Configuration, Configuring Dovecot for POP and IMAP C) Configuring Apache on Red Hat Enterprise Linux: Configuring the Apache Web Server, creating a Basic Website, Understanding the Apache Configuration Files, Apache Log Files, Working with Virtual Hosts, Securing the Web Server with TLS Certificates, Configuring Authentication, Setting Up Authentication with .htpasswd, Co	B) R, U, AN, A C) U, R, AN
V	<u>Bash Scripting, High-Availability Clustering, and Automated Installation on Red Hat Linux</u> A) Introducing Bash Shell Scripting: Introduction, Elements of a Good Shell Script, Executing the Script, Working with Variables and Input, Understanding Variables, Variables, Subshells, and Sourcing, Working with Script Arguments, Asking for Input, Using Command Substitution, Substitution Operators, Changing Variable Content with Pattern Matching, Performing Calculations, Using Control Structures, Using if...then...else, Using case, Using while, Using until, Using for, Configuring booting with GRUB. B) High-Availability Clustering: High-Availability Clustering, The Workings of High Availability, High-Availability Requirements, Red Hat High-Availability Add-on Software, Components, Configuring Cluster-Based Services, Setting Up Bonding, Setting Up Shared Storage, Installing the Red Hat High Availability Add-On, Building the Initial State of the Cluster, Configuring Additional Cluster Properties, Configuring a Quorum Disk, Setting Up Fencing, Creating Resources and Services, Troubleshooting a Nonoperational Cluster, Configuring GFS2 File Systems C) Setting Up an Installation Server: Configuring a Network Server as an Installation Server, Setting Up a TFTP and DHCP Server for PXE Boot, Installing the TFTP Server, Configuring DHCP for PXE Boot, Creating the TFTP PXE Server Content, creating a Kickstart File, Using a Kickstart File to Perform an Automated, Installation, Modifying the Kickstart File with, system-config-kickstart, Making Manual Modifications to the Kickstart File	A) R, U, A. AN B) R, U, AN, A, CR C) U, R, AN

List of Practical:		
Sr. No	Syllabus	Level of Knowledge Applicable as per Blooms Taxonomy
0	Installation of RHEL 6.X	U, A, AN, CR
1	Graphical User Interface and Command Line Interface and Processes	U, A, AN, CR
A	Exploring the Graphical Desktop	
B	The Command Line Interface	
C	Managing Processes	
2	Storage Devices and Links, Backup and Repository	U, A, AN
a	Working with Storage Devices and Links	
b	Making a Backup	
c	Creating a Repository	
3	Working with RPMsm Storage and Networking	U, A, AN, CR
A	Using Query Options	
B	Extracting Files From RPMs	
C	Configuring and Managing Storage	
D	Connecting to the Network	
4	Working with Users, Groups, and Permissions	U, A, AN, CR
5	Firewall and Cryptographic services	U, A, AN
A	Securing Server with iptables	
B	Setting Up Cryptographic Services	
6	Configuring Server for File Sharing	U, A, AN

A	Configuring NFS Server and Client	
B	Configuring Samba	
C	Configuring FTP	
7	DNS, DHCP and Mail Server	U, A, AN, CR
A	Configuring DNS	
B	Configuring DHCP	
C	Setting Up a Mail Server	
8	Web Server	U, A, AN, CR
A	Configuring Apache on Red Hat Enterprise Linux	
B	Writing a Script to Monitor Activity on the Apache Web Server	
C	Using the select Command	
9	Shell Scripts and High-Availability Clustering	U, A, AN, CR
A	Writing Shell Scripts	
B	Configuring Booting with GRUB	
C	Configuring High Availability Clustering	
10	Setting Up an Installation Server	U, A, AN, CR
A	Configuring Network Server as an Installation Server	
B	Setting Up a TFTP and DHCP Server for PXE Boot	

Learning Outcomes: After completion of Course, the learners will be able to:

1. Gain proficiency in Red Hat Enterprise Linux administration, including working with the Bash shell, performing file system management tasks, and understanding basic system administration duties.
2. Develop skills in storage configuration, networking, and service management. Learn to configure and manage logical volumes, snapshots, encrypted volumes, and network services.
3. Acquire knowledge and practical experience in user and group management, including password management, external authentication, permissions, and access control lists.
4. Implement server security measures using iptables, including firewall configuration, advanced settings, and cryptographic services for SSL and GPG.
5. Demonstrate proficiency in configuring and managing server features, including file sharing with NFS and Samba, DNS and DHCP services, mail server setup, Apache web server configuration, and Bash shell scripting.

Books and References:

1. van Vugt, S. (2013). Red Hat Enterprise Linux 6 administration. John Wiley & Sons.
2. Collings, T., & Wall, K. (n.d.). Red Hat Linux networking and system administration (3rd ed.). Wiley.
3. Soyinka, W. (n.d.). Linux administration: A beginner's guide (5th ed.). TMH.
4. Volkerding, P. (2018). The Linux cookbook (2nd ed.). No Starch Press.
5. Negus, C. (2019). Linux Bible (9th ed.). Wiley.
6. Blum, R. (2015). Linux command line and shell scripting Bible (4th ed.). Wiley.
7. Sobell, M. G. (2017). A practical guide to Linux commands, editors, and shell programming (4th ed.). Pearson Education.
8. Negus, C. (2020). Linux Bible (10th ed.). Wiley.
9. Ward, B. (2014). How Linux works: What every superuser should know (2nd ed.). No Starch Press.
10. Purcell, M. (2020). Linux for beginners: An introduction to the Linux operating system and command line. Independently Published.

Percentage of 6 categories of Blooms Taxonomy in question paper

	Remember	Understand	Apply	Analyze	Evaluate	Create	
% in Question Paper	20	20	20	15	10	15	100%

MCCSCTPRJ402 CAPSTONE Project

B. Sc. (Information Technology)		Semester – VI	
Course Name: CAPSTONE Project		Course Code: MCCSCTPRJ402	
Vertical		RP / OJT	
Periods per week (1 Period is 50 minutes)		3	
Credits		4	
		Hours	Marks
Evaluation System	Presentation & Viva	3	100

The details are given in CAPSTONE Project Policy document.

Evaluation Scheme

The Exam for the students in this programme will be held under four heads

- Continuous Internal Assessment
- End Semester Examination
- Practical Continuous Internal Examination
- Practical End Semester Examination

1. Internal Evaluation (40 Marks).

i. Test: 1 and Test 2: Class test of 20 marks. (Can be taken online)

Q	Attempt <u>any four</u> of the following:	20
a.		
b.		
c.		
d.		
e.		
f.		

2. End Semester Examination: (60 marks)

	All questions are compulsory	
Q1	(Based on Unit 1) Attempt <u>any two</u> of the following:	12
a.		
b.		
c.		
d.		
Q2	(Based on Unit 2) Attempt <u>any two</u> of the following:	12
Q3	(Based on Unit 3) Attempt <u>any two</u> of the following:	12
Q4	(Based on Unit 4) Attempt <u>any two</u> of the following:	12
Q5	(Based on Unit 5) Attempt <u>any two</u> of the following:	12

3. End Semester Examination: (30 marks)

	All questions are compulsory	
Q1	(Based on Unit 1) Attempt <u>any one</u> of the following:	6
a.		
b.		
Q2	(Based on Unit 2) Attempt <u>any one</u> of the following:	6
Q3	(Based on Unit 3) Attempt <u>any one</u> of the following:	6
Q4	(Based on Unit 4) Attempt <u>any one</u> of the following:	6
Q5	(Based on Unit 2) Attempt <u>any one</u> of the following:	6

4. Practical End Semester Examination

Note: Certified Copy of Journal is mandatory for appearing for the practical exam.

1.	Practical Question 1	15
2.	Practical Question 2	15

OR

1.	Practical Question	30
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For Tutorial Exam, a paper of 30 marks to be solved.

- 5. General Elective / Open Elective will have continuous internal assessment of 50 marks. The assessment shall be based on Assignments / Tests / Presentations/ Role plays and similar activities.**